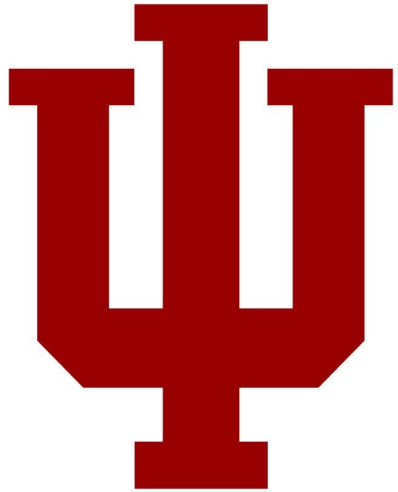
01: Special Signals

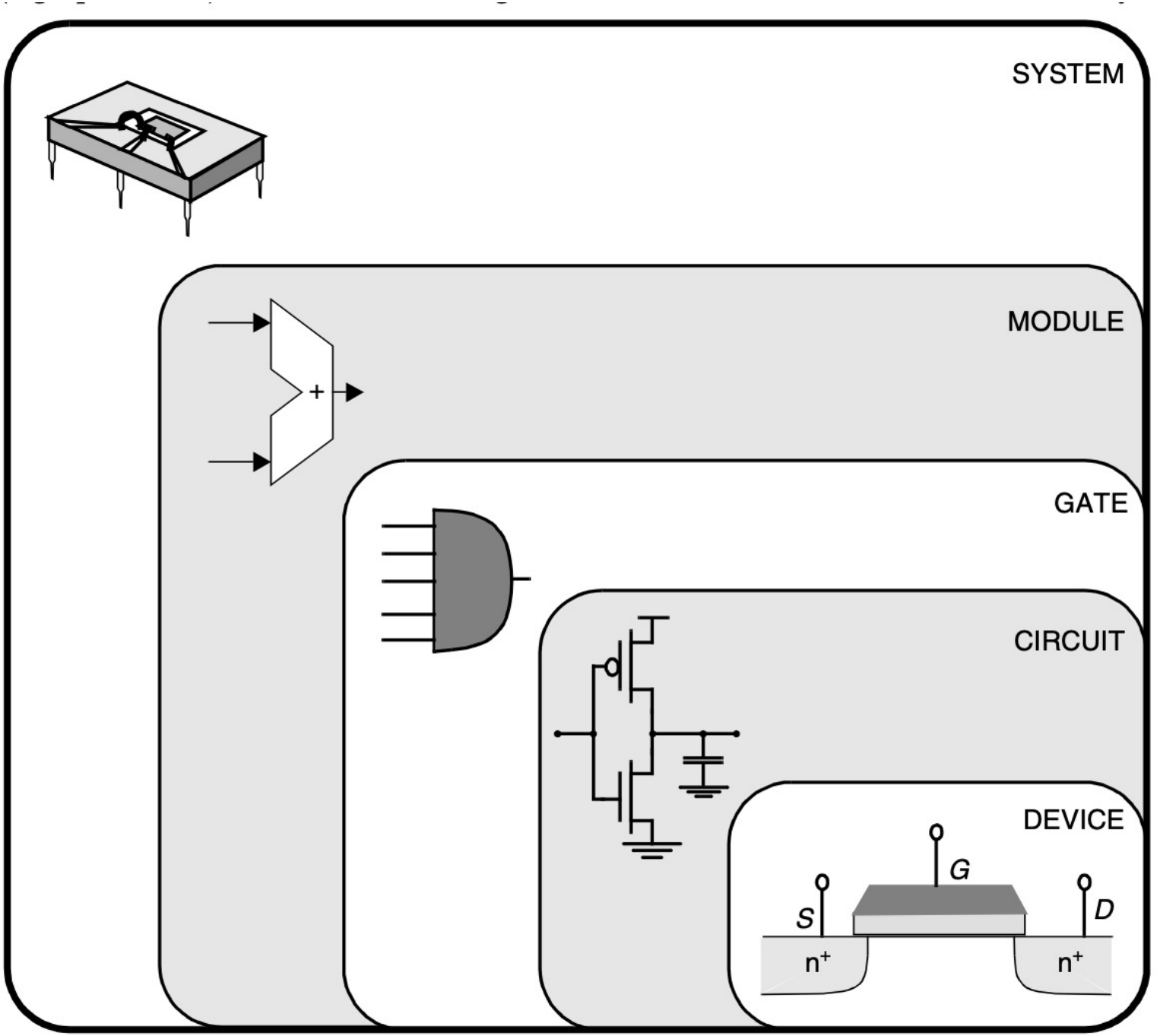
E599– Digital Integrated Circuit Design



Website

https://engr599-ic.github.io

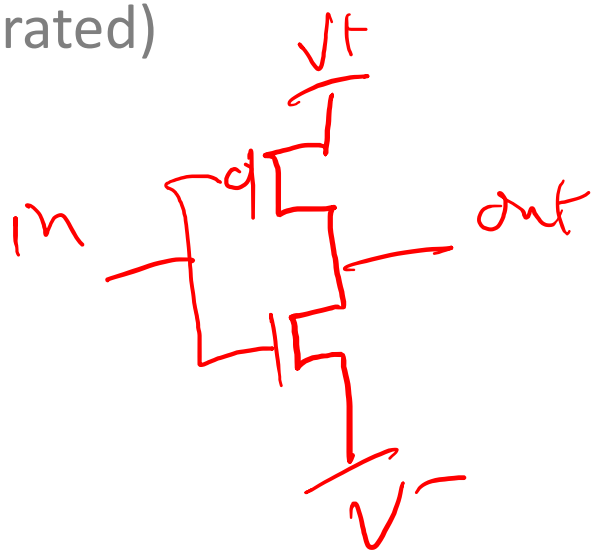
Review Time



[Digital Integrated Circuits: A Design Perspective]

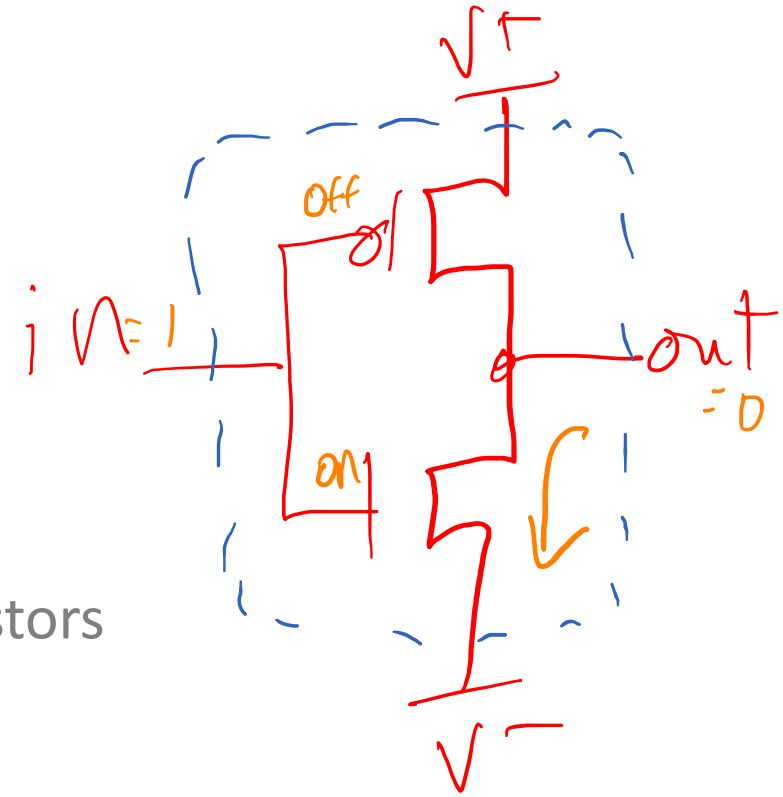
Devices Review

- Transistors
  - Voltage on gate causes conduction between source + drain. Aka “Magic”
  - Only consider fully-off or fully-on (saturated)
  - Two types: NMOS and PMOS
    - PMOS: 0V=on
    - NMOS: 0V =off



- Wires
  - Resistor + Capacitor
  - Longer wires = slower signals

Combinational Circuits Review



- Inverter: 1 NMOS + 1 PMOS
- Process Variation:
  - Some faster, some slower transistors
- Intrinsic Delay:
  - Fundamental delay in all transitions
  - Bigger transistors reduce delays, to a point.
- Other Delays
  - output takes times to reflect input changes
  - Bigger capacitive loads increase delays

fanout:

Sequential Circuits Review

- Latches: Output follows input when clock *is high*
- Flip-Flop: Output follows input ***on the rising edge*** of the clock

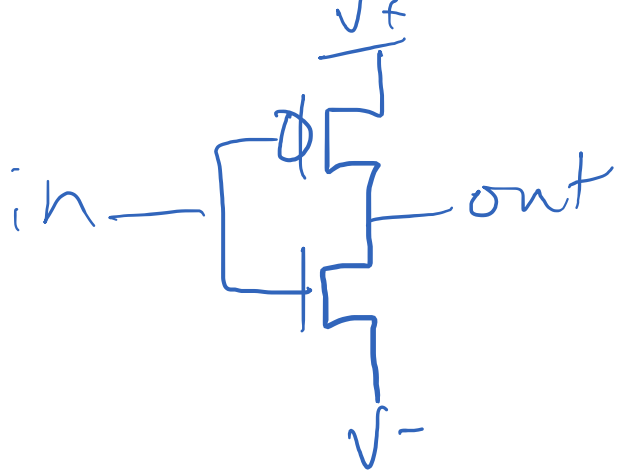
- Setup Time: Time input must be stable **before** the clock rising edge
- Hold Time: Time input must be stable **after** the clock rising edge

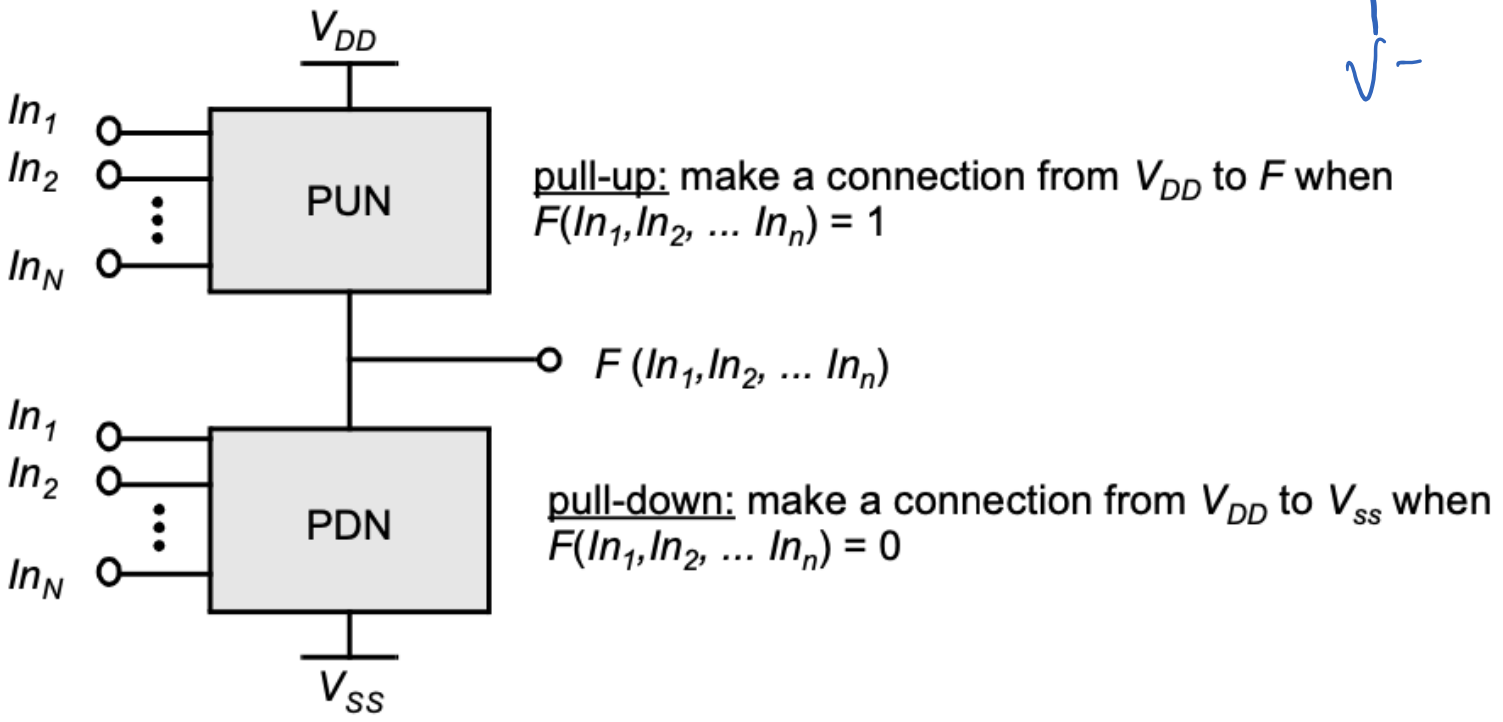
- Setup affects performance. **Hold affects correctness**.

WARNING: HOLD TIMES ARE A BIG DEAL!!!

- You can always fix setup times with a slower clock.
- Hold times are unfixable….
- You do not want a hold-time violation in your design!

Standardized Logic Gates



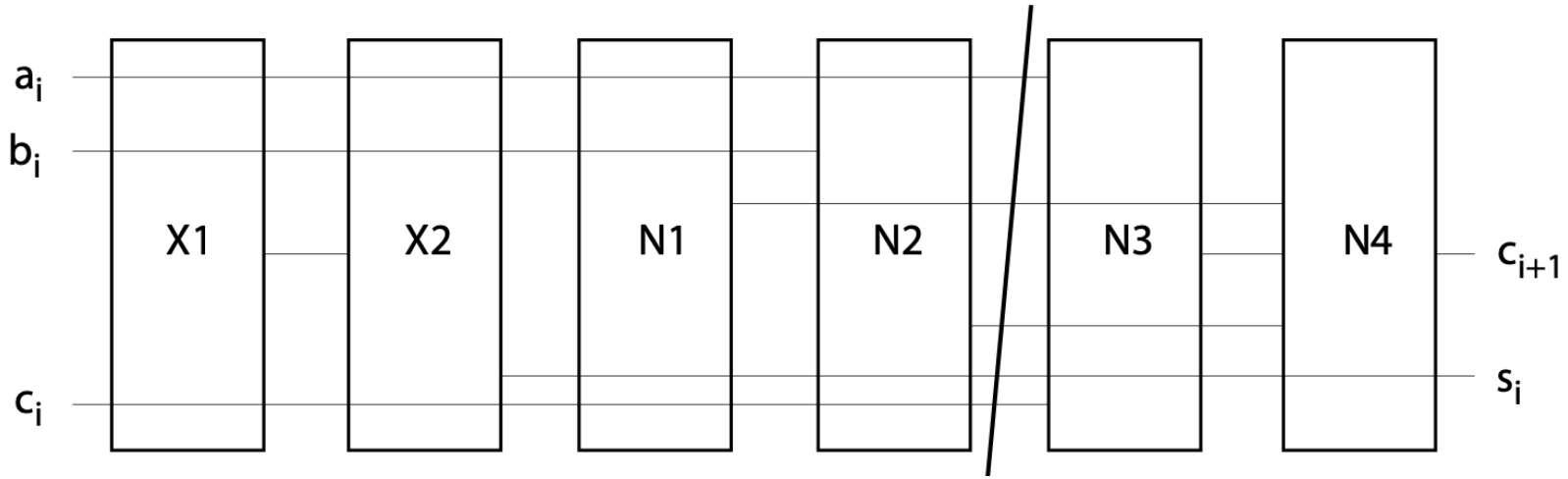


pull-up: make a connection from V_{DD} to F when $F(In_1, In_2, \ldots\ In_n) = 1$

pull-down: make a connection from V_{DD} to V_{ss} when $F(In_1, In_2, \ldots\ In_n) = 0$

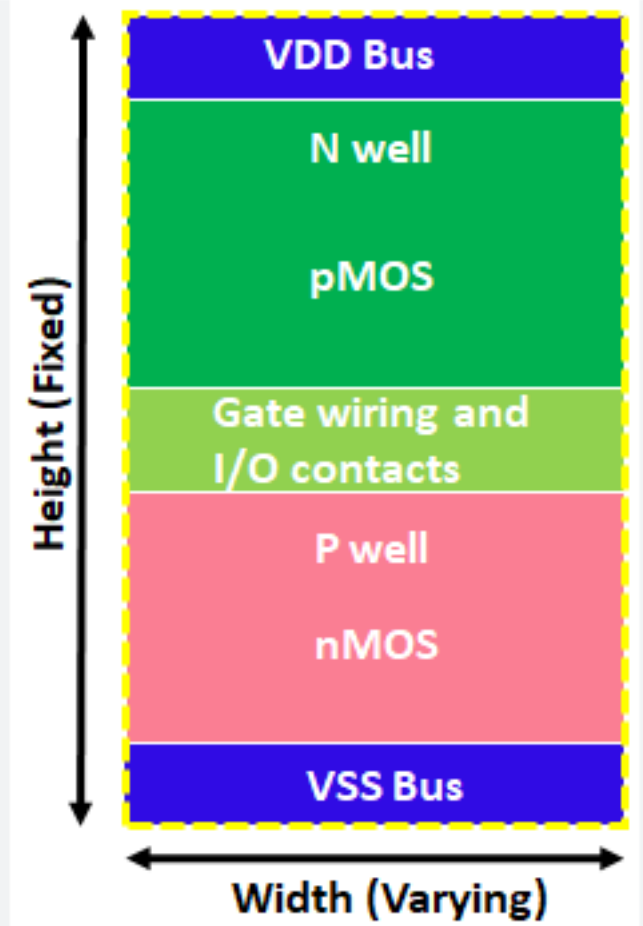
[Digital Integrated Circuits: A Design Perspective]

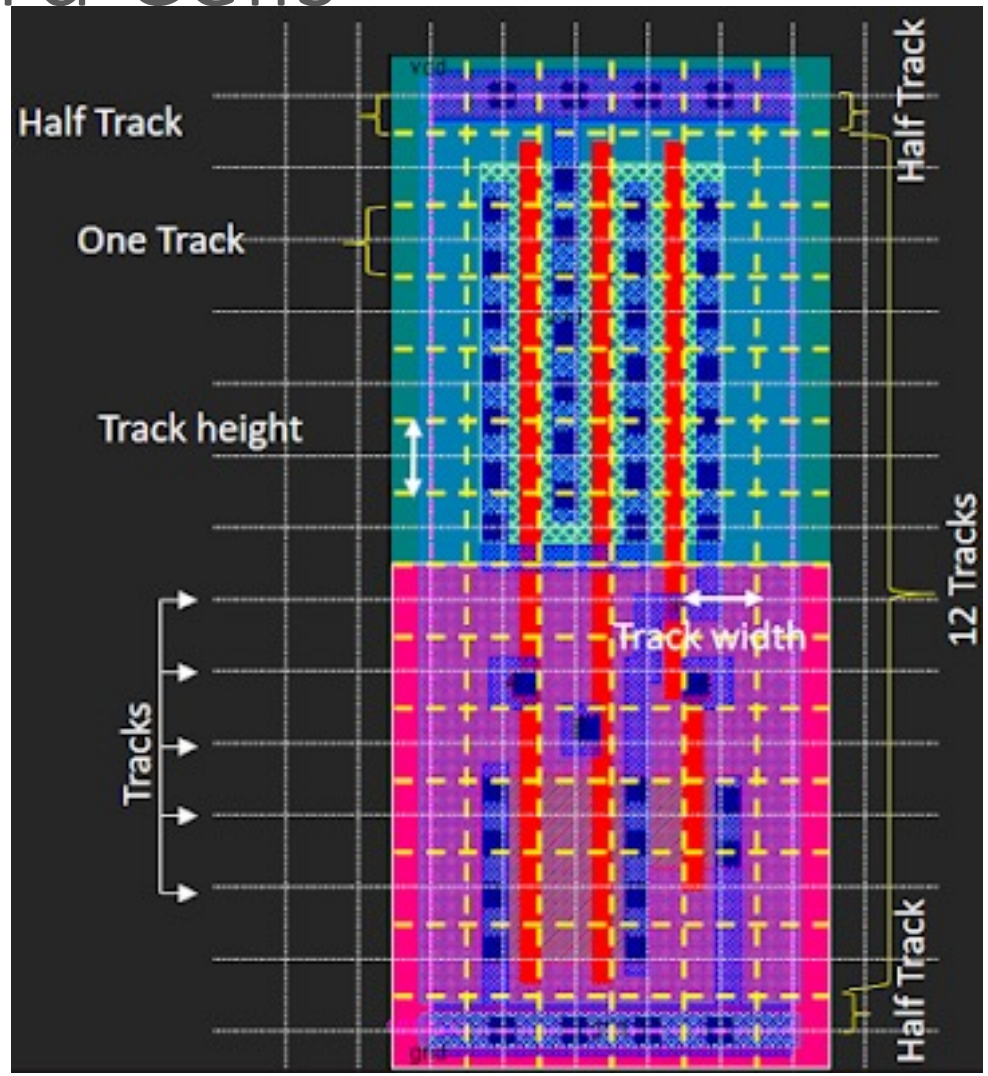
Put the wires above the logic cells



[***Modern VLSI Design: IP-Based Design***]

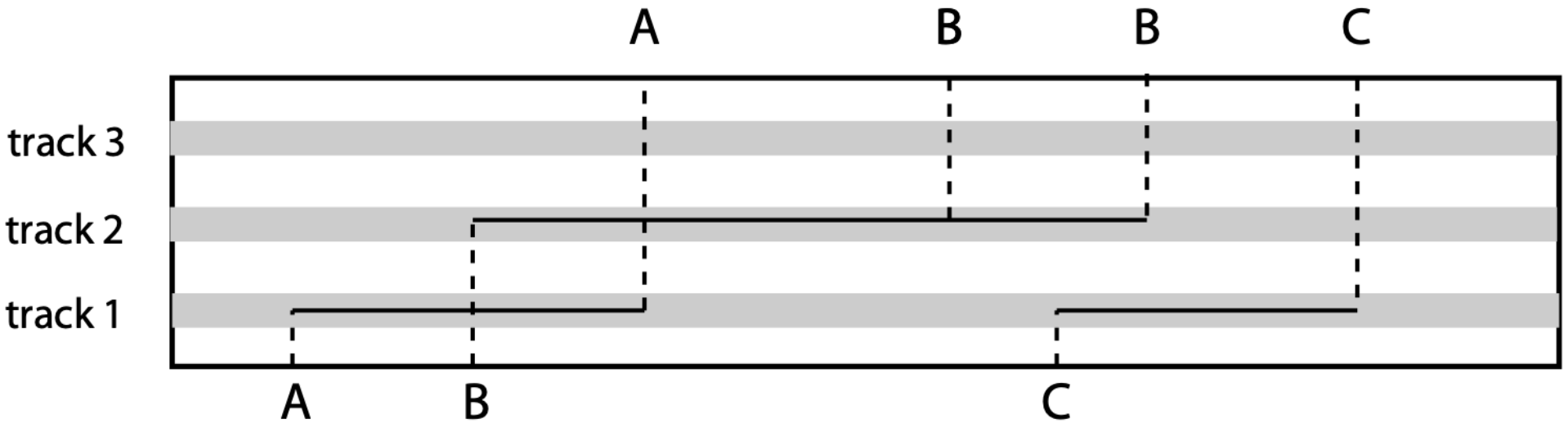
Routing Tracks w/in Standard Cells





https://teamvlsi.com/2020/05/standard-cells-in-asic-design-standard-cells-in-vlsi.html

Routing Tracks



[***Modern VLSI Design: IP-Based Design***]

The “CLK” (Clock)

- In digital logic, a clock signal is a periodic electrical signal, typically a square wave, that oscillates between high and low states to provide a precise timing reference for synchronous circuits [wiki]

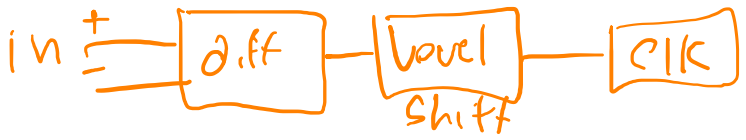
Getting the CLK

- Where does CLK come from?

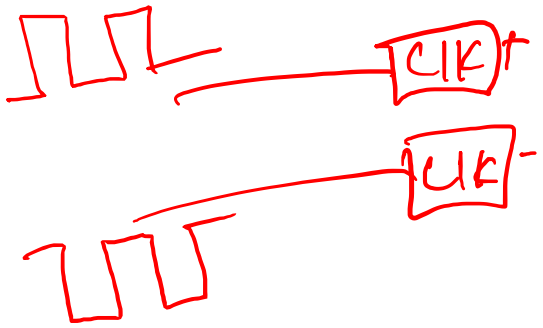
- CLK input
- Oscillator
  - crystal
  - "voltage controlled oscillator"
  - "Phase Locked Loop"
  - "Delay Locked Loop"
- RC circuits

Getting the CLK

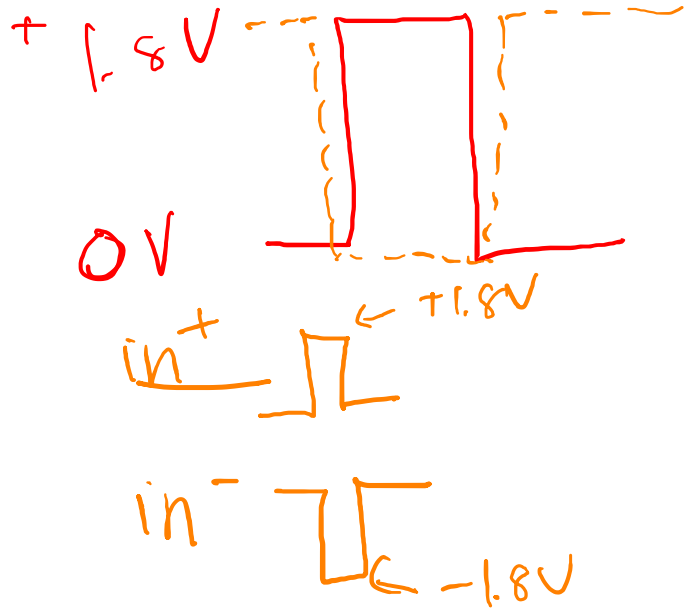
- Where does CLK come from?
- Outside world
  - Limited frequency to <200MHz
- Inside IC
  - Requires additional hardware
  - Often requires outside reference clock

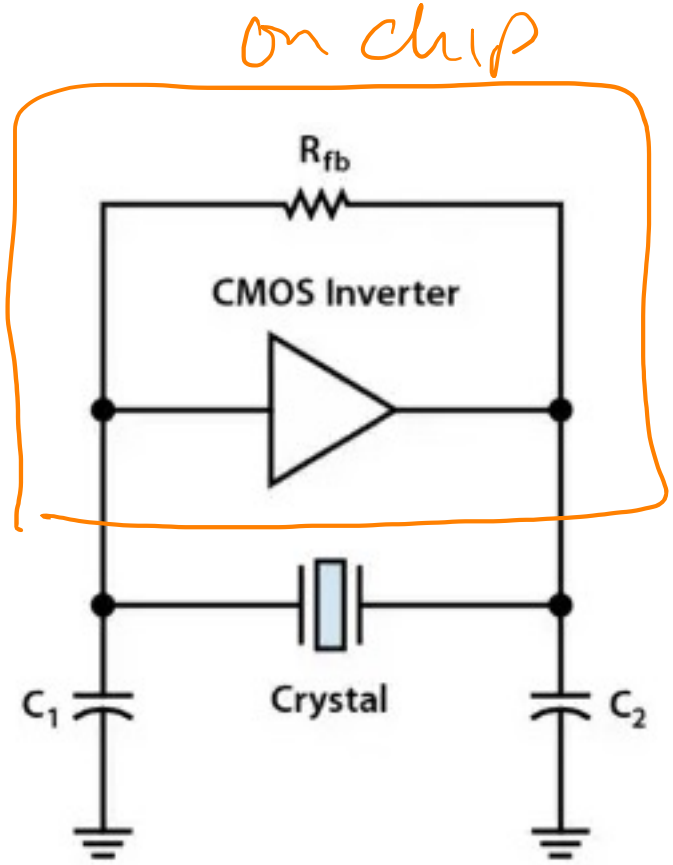


CLKs from the outside world



- Single-ended signal
- Differential signal
- Crystal Oscillator

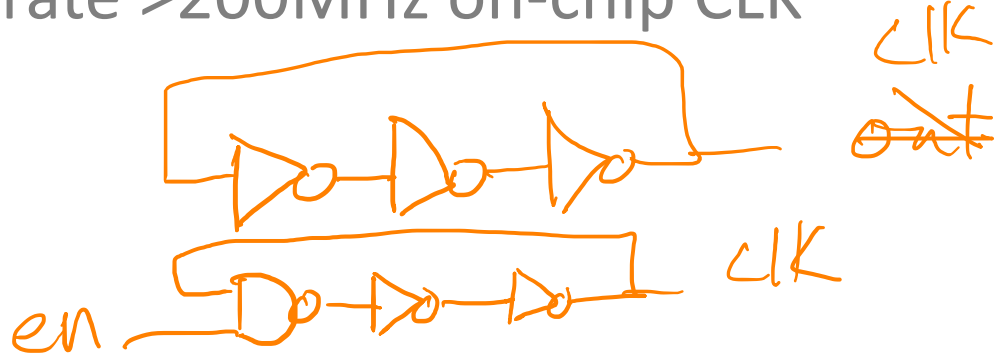




https://www.electrical4u.com/crystal-oscillator/

CLKs from inside the IC

- Need a block to generate >200MHz on-chip CLK



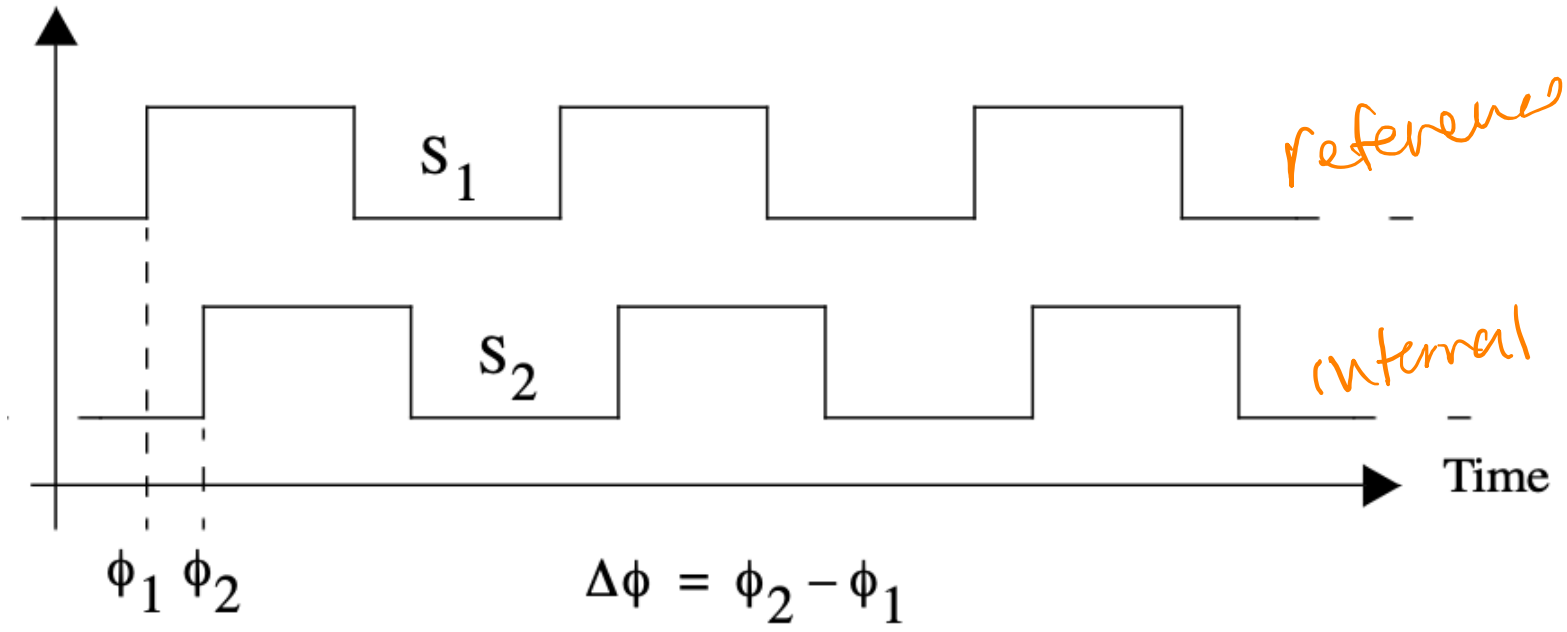
- Lots of options
  - Ring Oscillator
  - RC (LC) oscillator
  - **Phase-Locked Loop (PLL)**
  - Delay-Locked Lopp (DLL)

slower requires reference freq. CLK

PLLs

- Basics:
  - Input: Low-frequency off-chip “reference” clock
  - Magic
  - Output: high-frequency on-chip clock

PLL Basics



[Digital Integrated Circuits: A Design Perspective]

PLL Basics

Voltage controlled oscillator

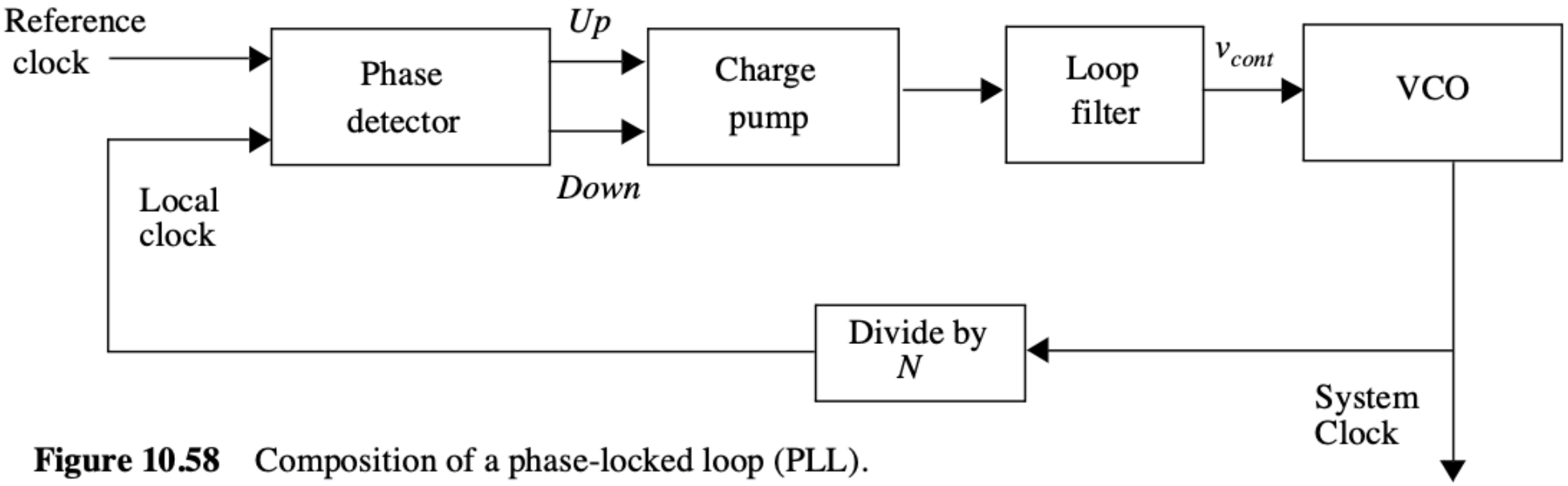
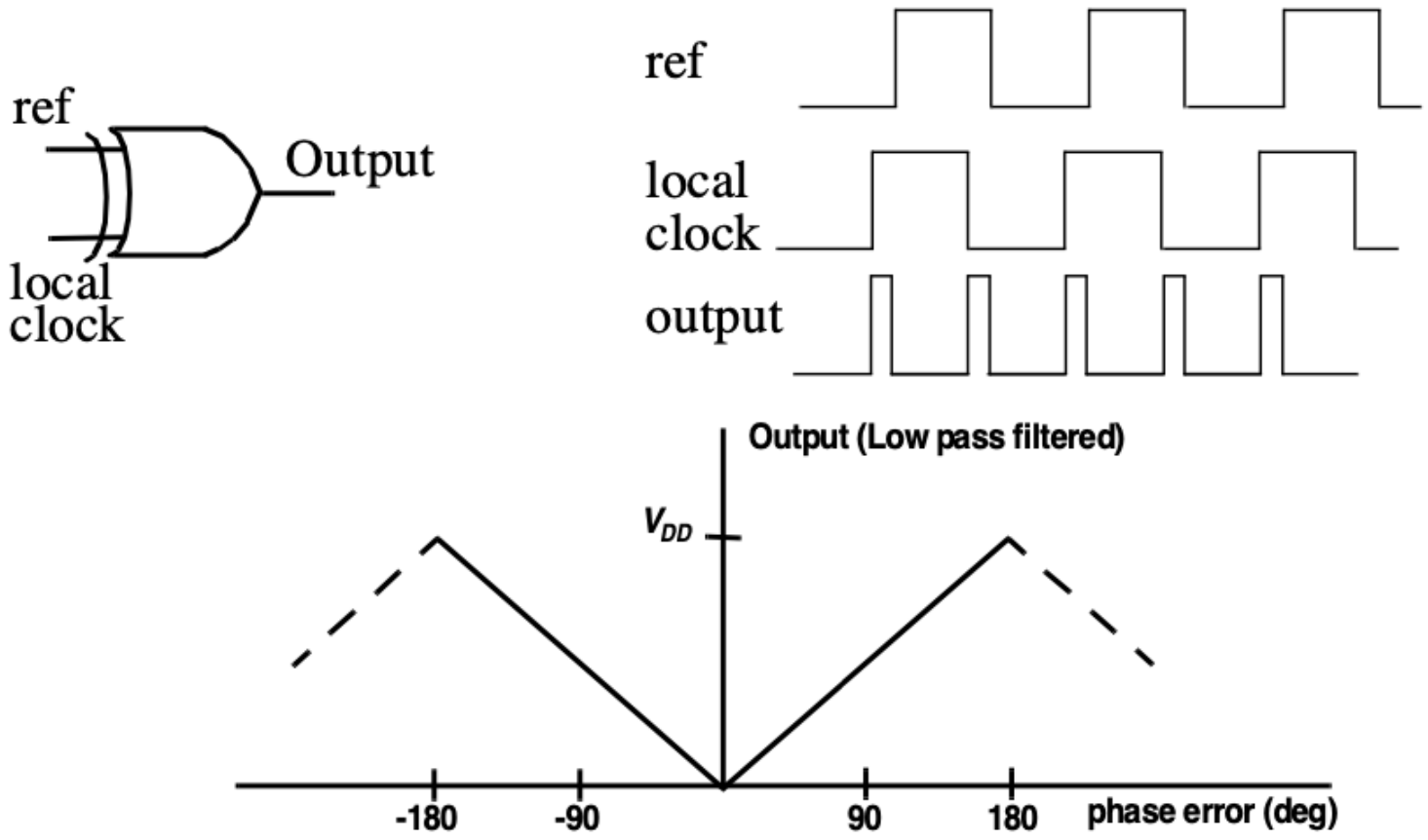


Figure 10.58 Composition of a phase-locked loop (PLL).

[Digital Integrated Circuits: A Design Perspective]

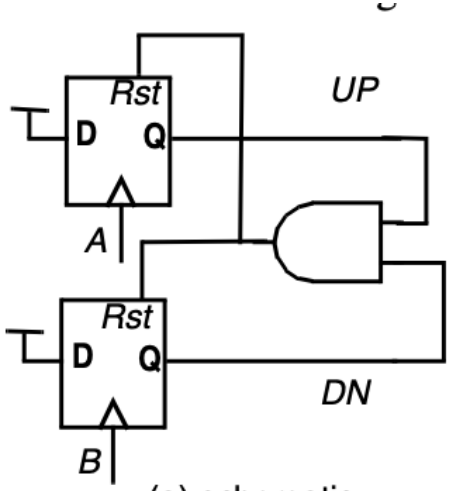
PLL Basics



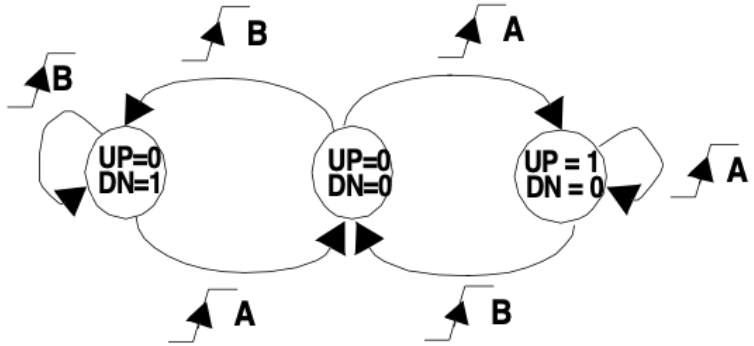
Can lock onto the wrong multiple of clock frequency

[Digital Integrated Circuits: A Design Perspective]

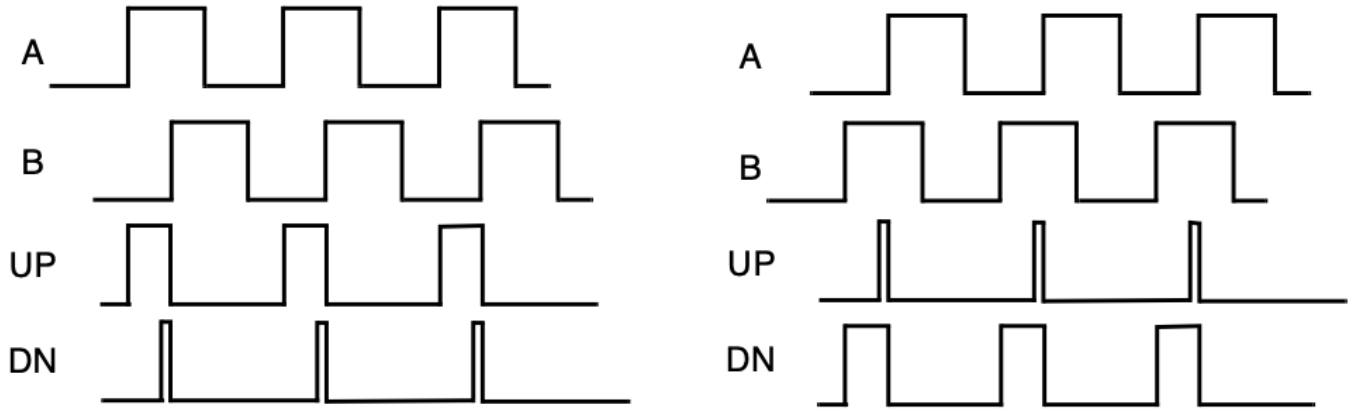
PLL Basics



(a) schematic



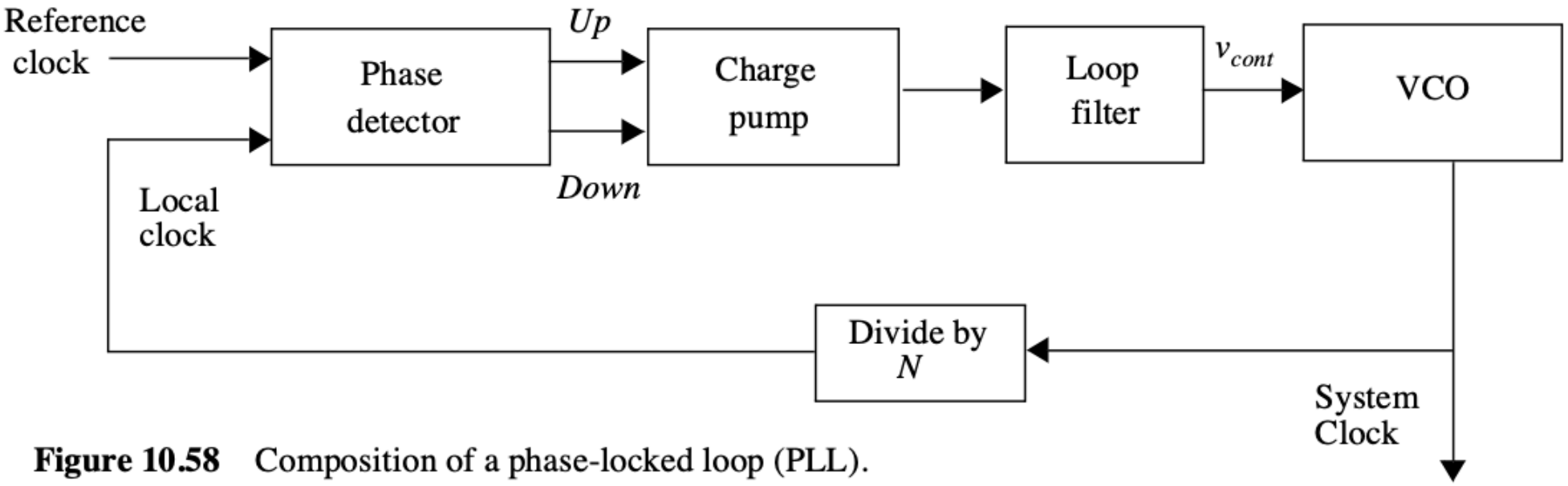
(b) state transition diagram



(c) Timing waveforms

[Digital Integrated Circuits: A Design Perspective]

PLL Basics



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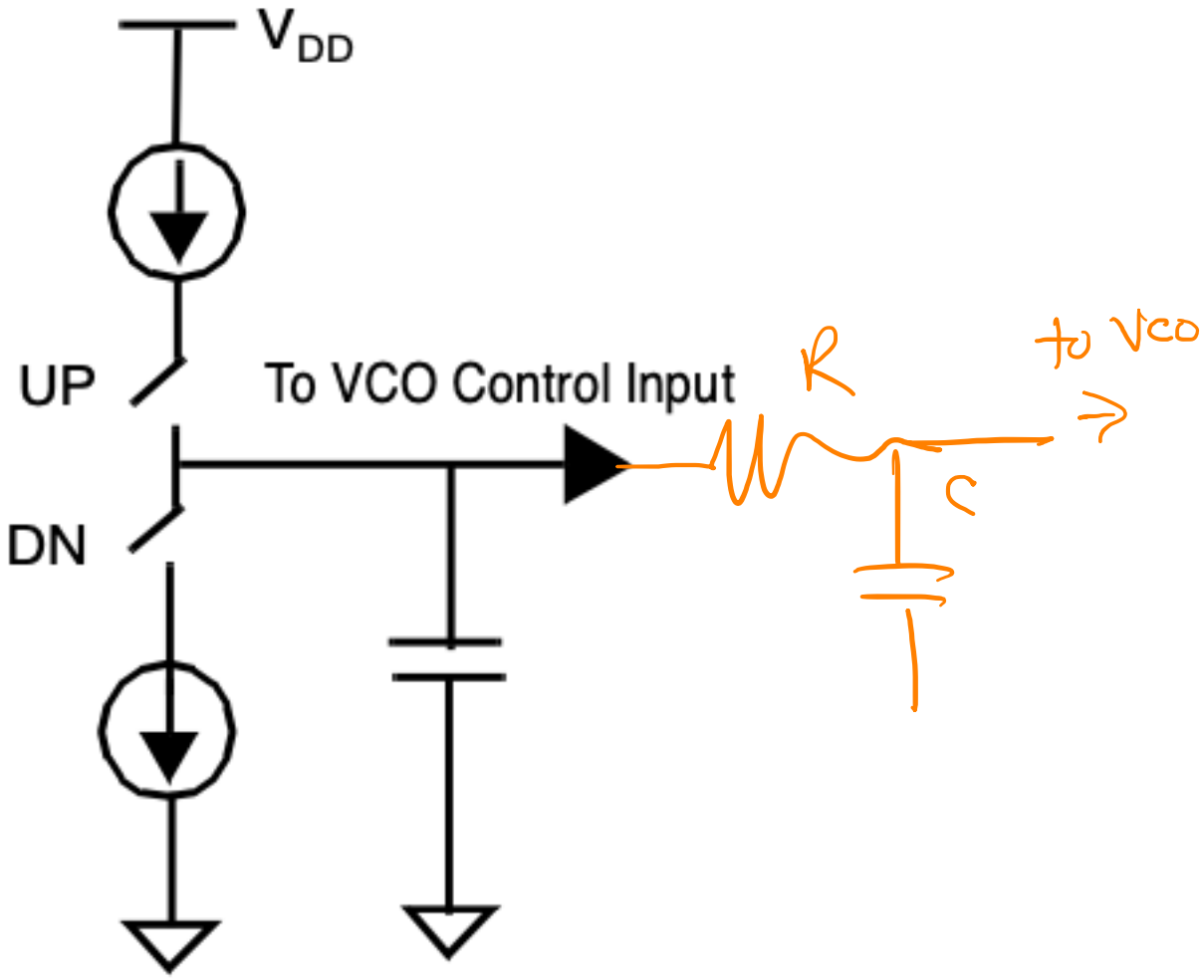
[Digital Integrated Circuits: A Design Perspective]

PLL Basics



[Digital Integrated Circuits: A Design Perspective]

PLL Basics

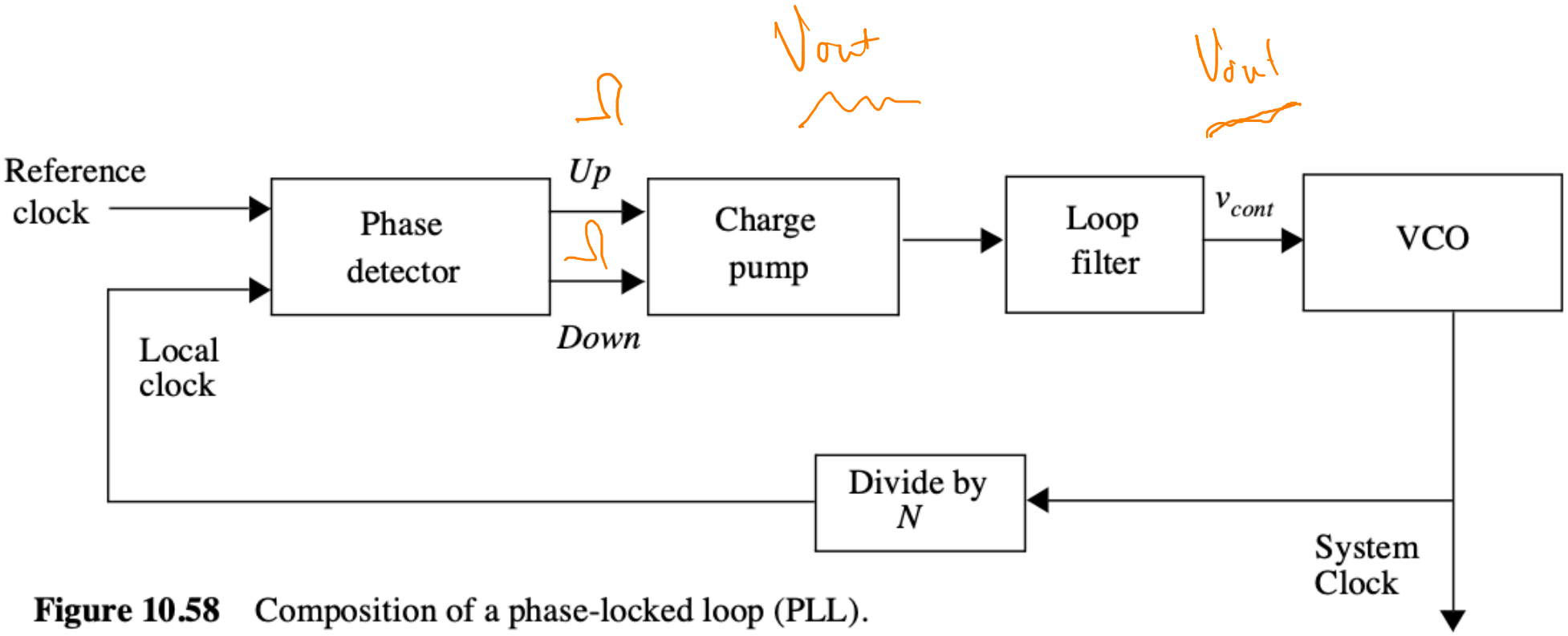
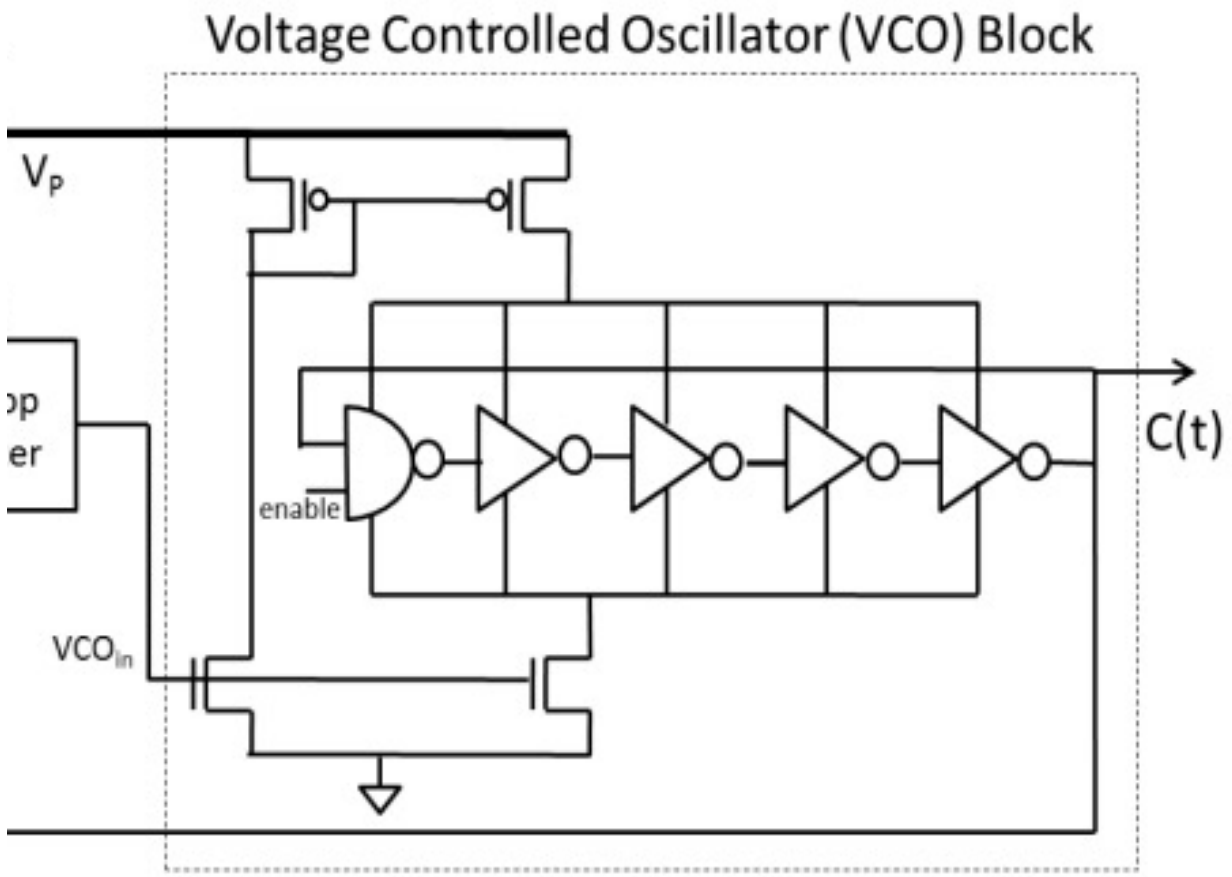


Figure 10.58 Composition of a phase-locked loop (PLL).

[Digital Integrated Circuits: A Design Perspective]

PLL Basics



https://www.researchgate.net/figure/PLL-circuit-with-the-conventional-ring-oscillator-type-VCO_fig6_303307684

PLL Basics

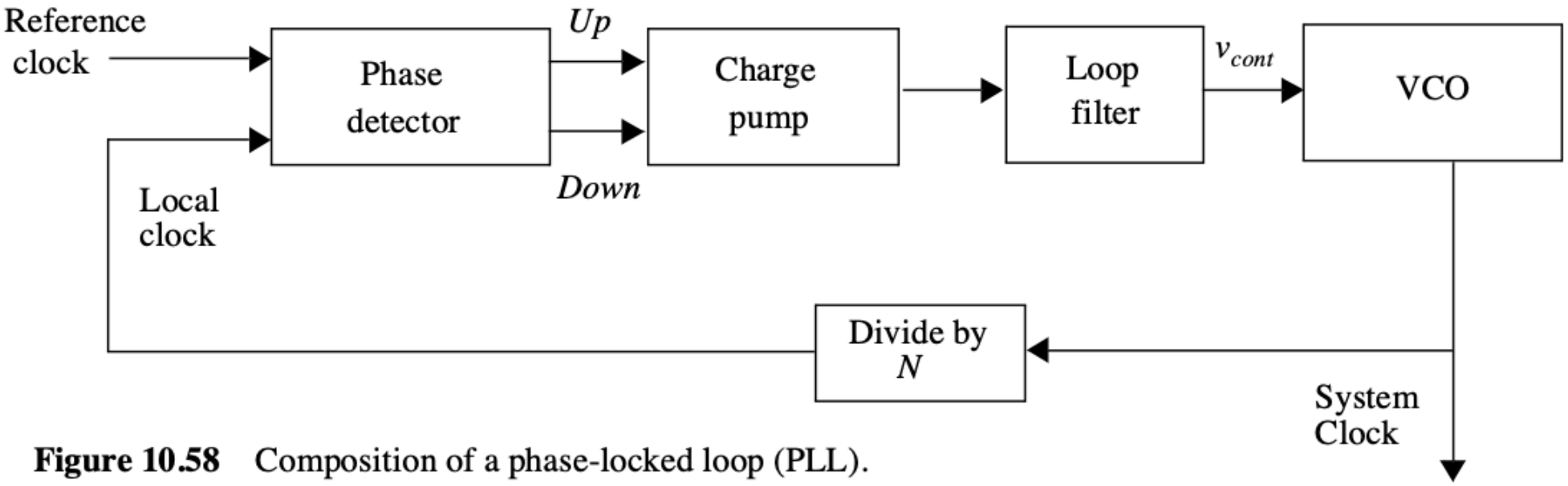
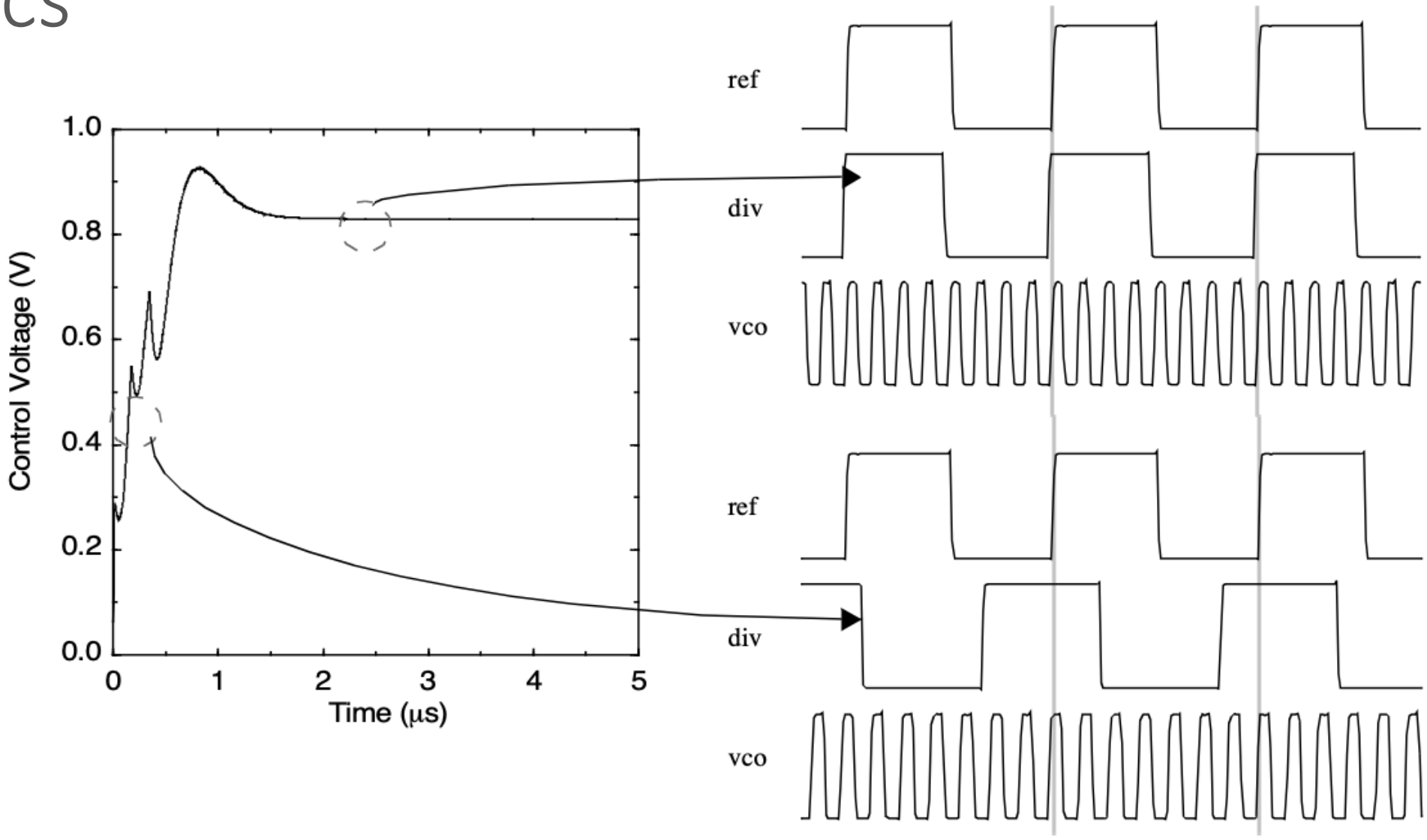


Figure 10.58 Composition of a phase-locked loop (PLL).

[Digital Integrated Circuits: A Design Perspective]

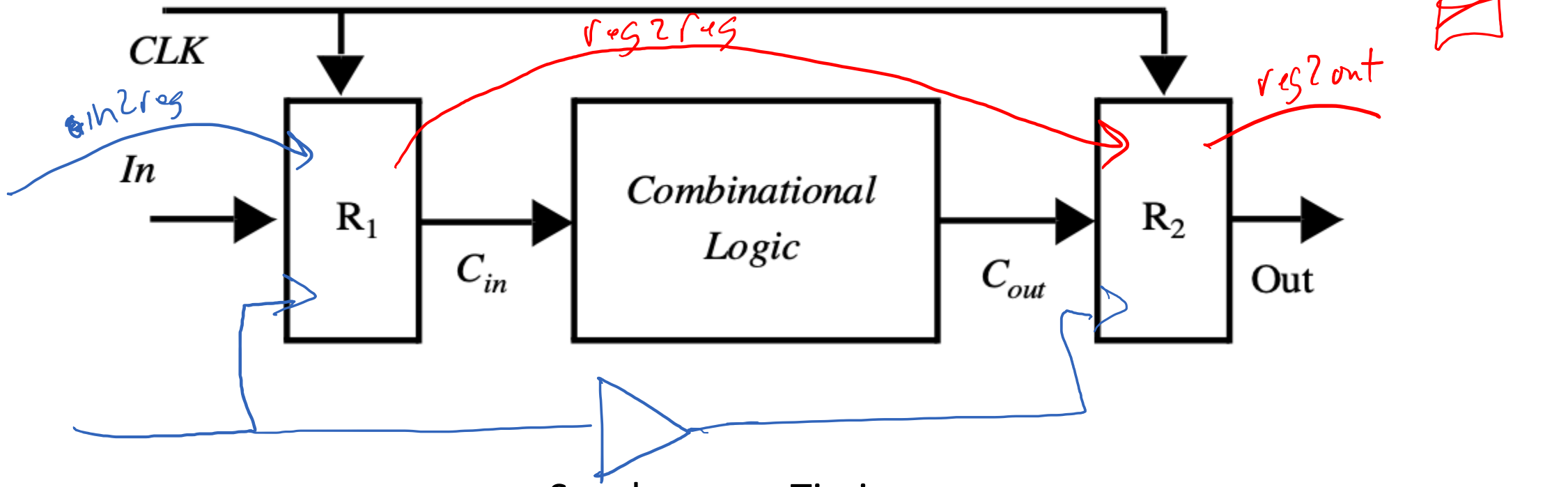
PLL Basics



[Digital Integrated Circuits: A Design Perspective]

Different Timing Arcs

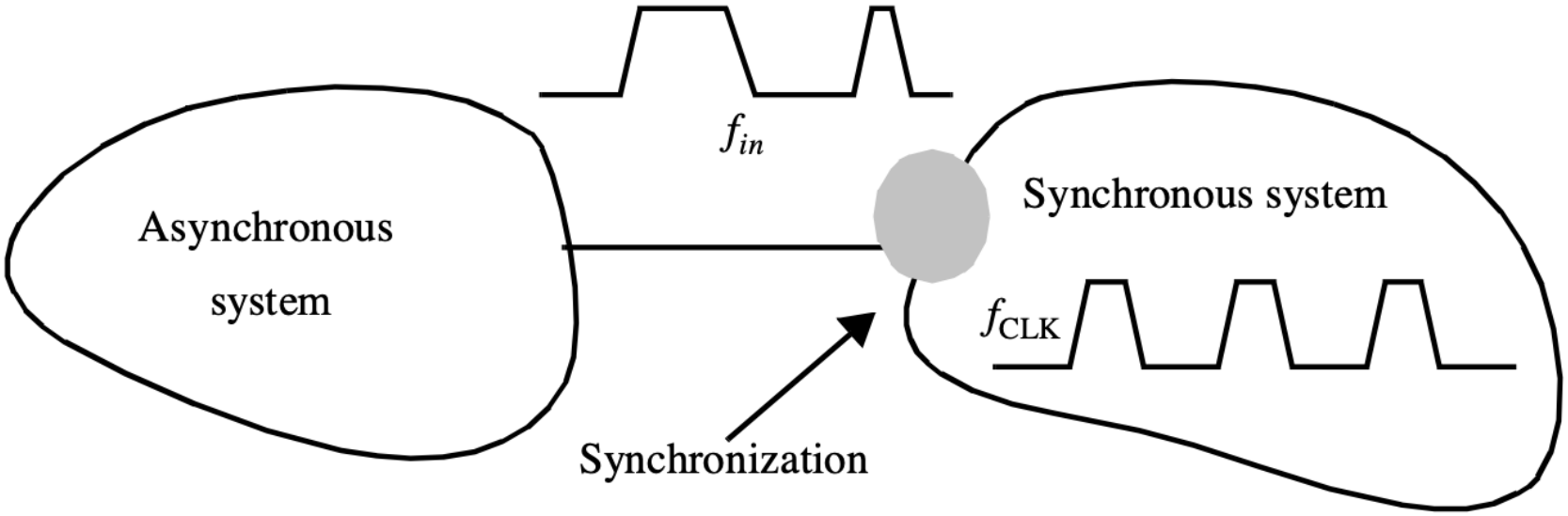
Reg2reg
In2reg
reg2out



Synchronous Timing

[Digital Integrated Circuits: A Design Perspective]

How do we deal with asynchronous inputs?



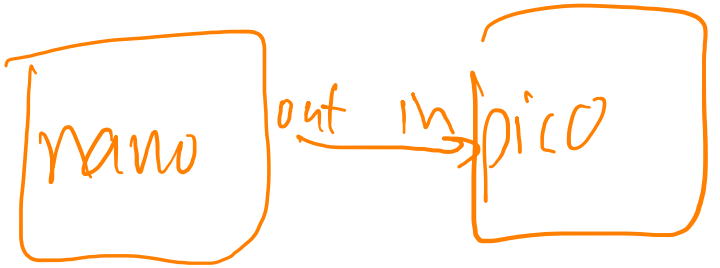
[Digital Integrated Circuits: A Design Perspective]

We force them to be synchronous

- Use SDCs to define input/output timing constraints
- Works well for blocks.

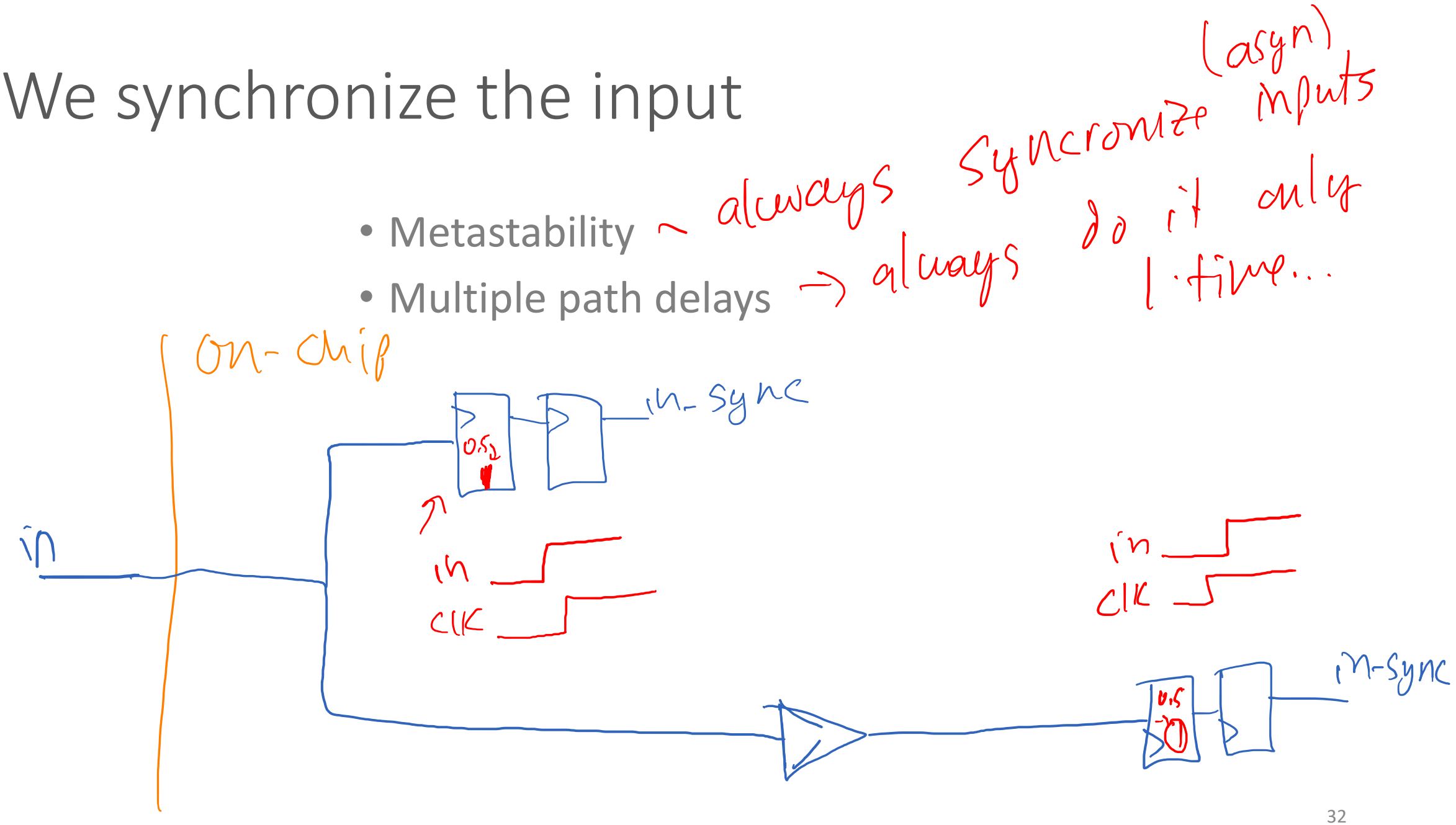
```
set_input_delay -clock <clock_name>
[-max] [-min] <delay_value>
[get_ports <port_list>]
```


```
create_clock -name clk -period 500 [get_ports {clk}]

set_input_delay -clock [get_clocks clk] -min -add_delay 100 [all_inputs]
set_input_delay -clock [get_clocks clk] -max -add_delay 400 [all_inputs]

set_output_delay -clock [get_clocks clk] -min -add_delay 2 [all_outputs]
set_output_delay -clock [get_clocksclk] -max -add_delay  400 [all_outputs]
```

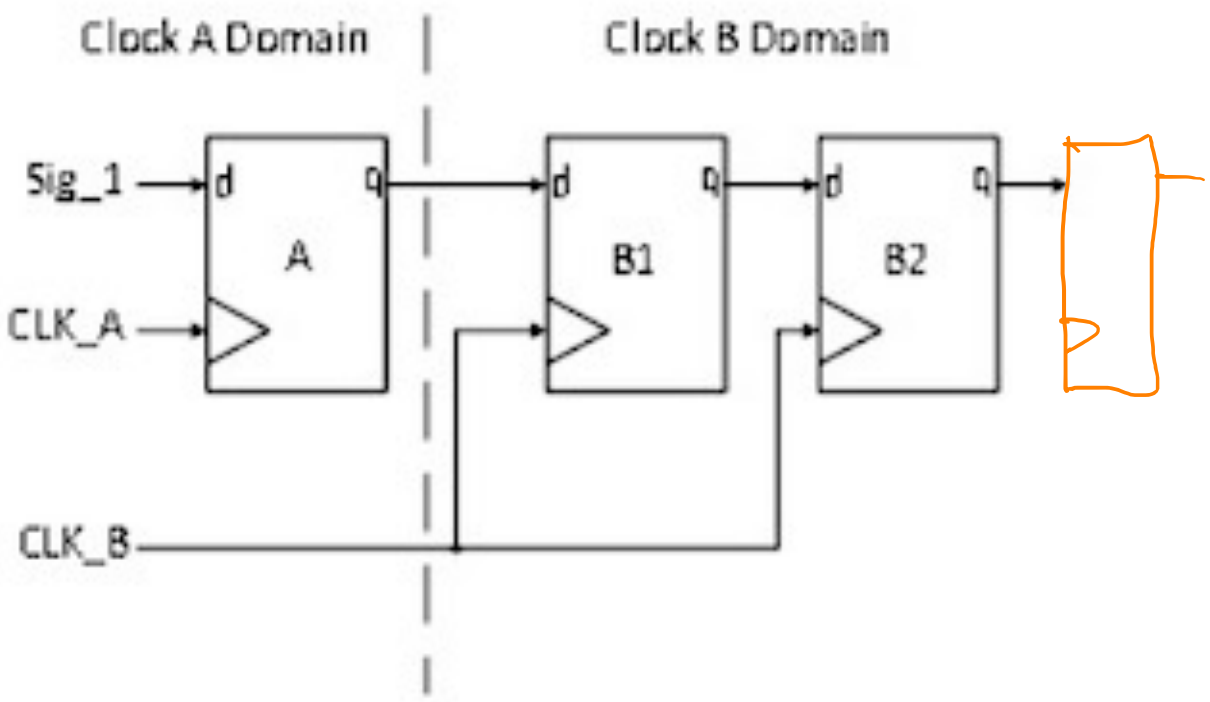


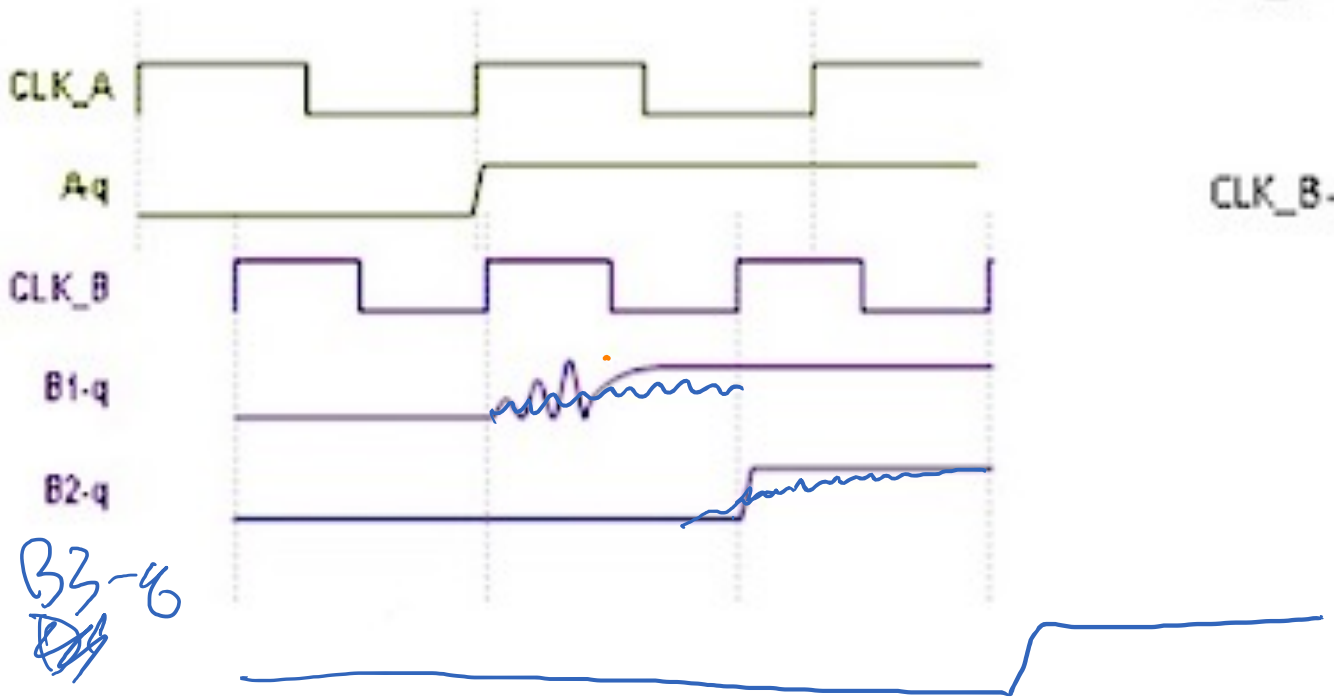
We synchronize the input

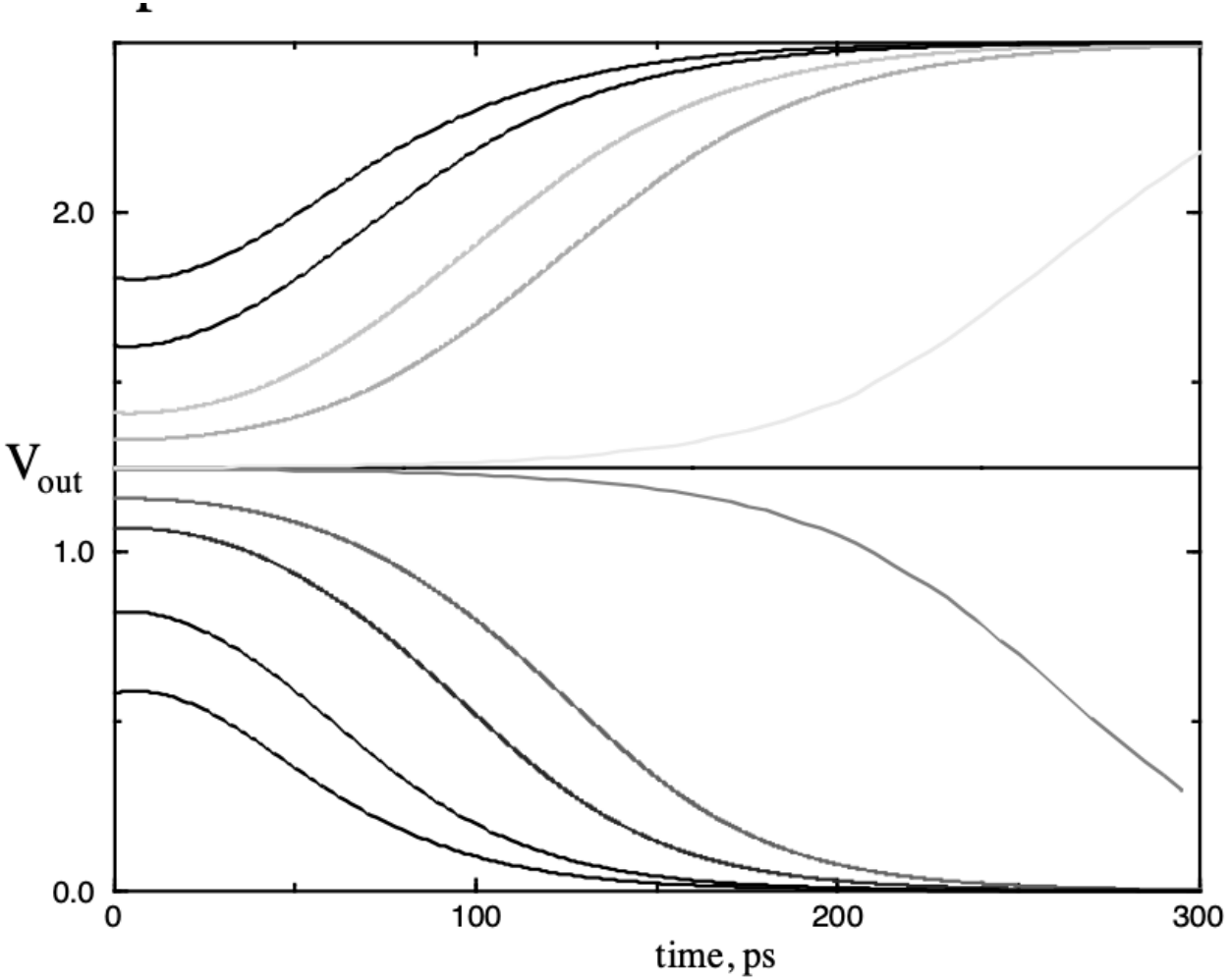
- Metastability
- Multiple path delays

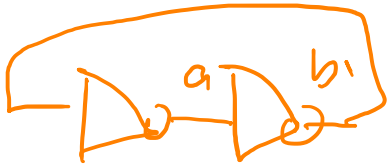


We synchronize the input





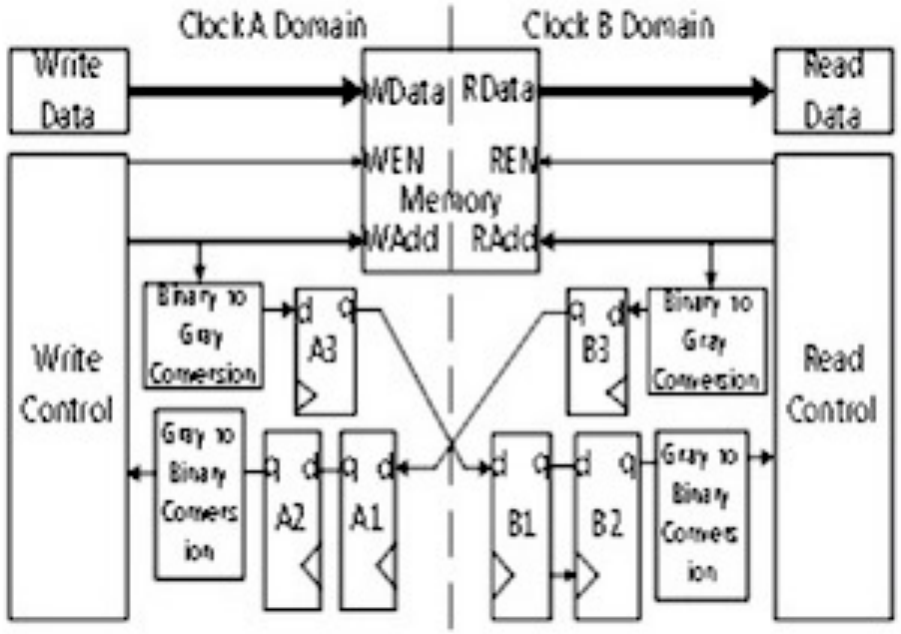


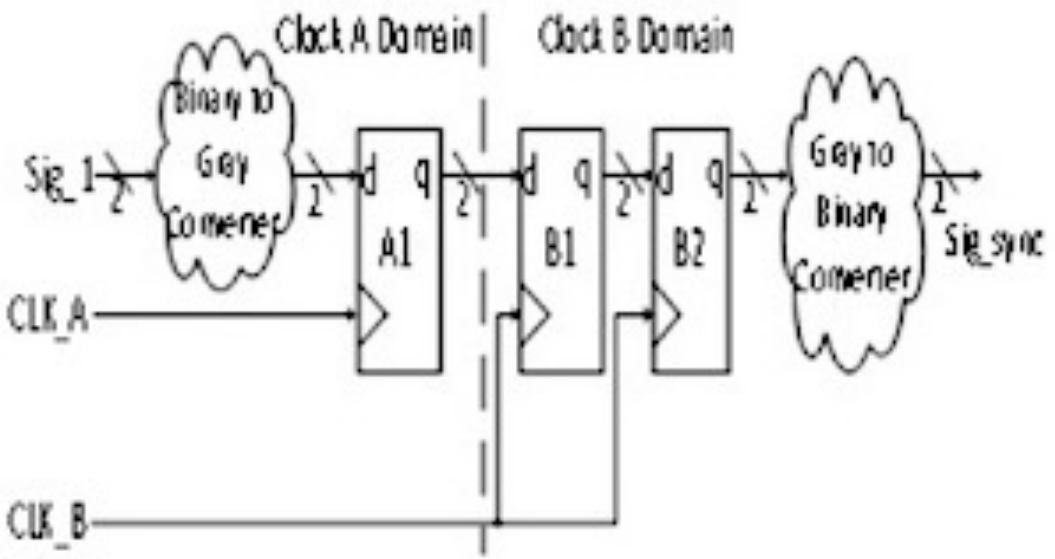


a b
0.4 0.65
0.3 → 0.75
0.2 → 0.85
0.1 → 0.95
0.0

[Digital Integrated Circuits: A Design Perspective]

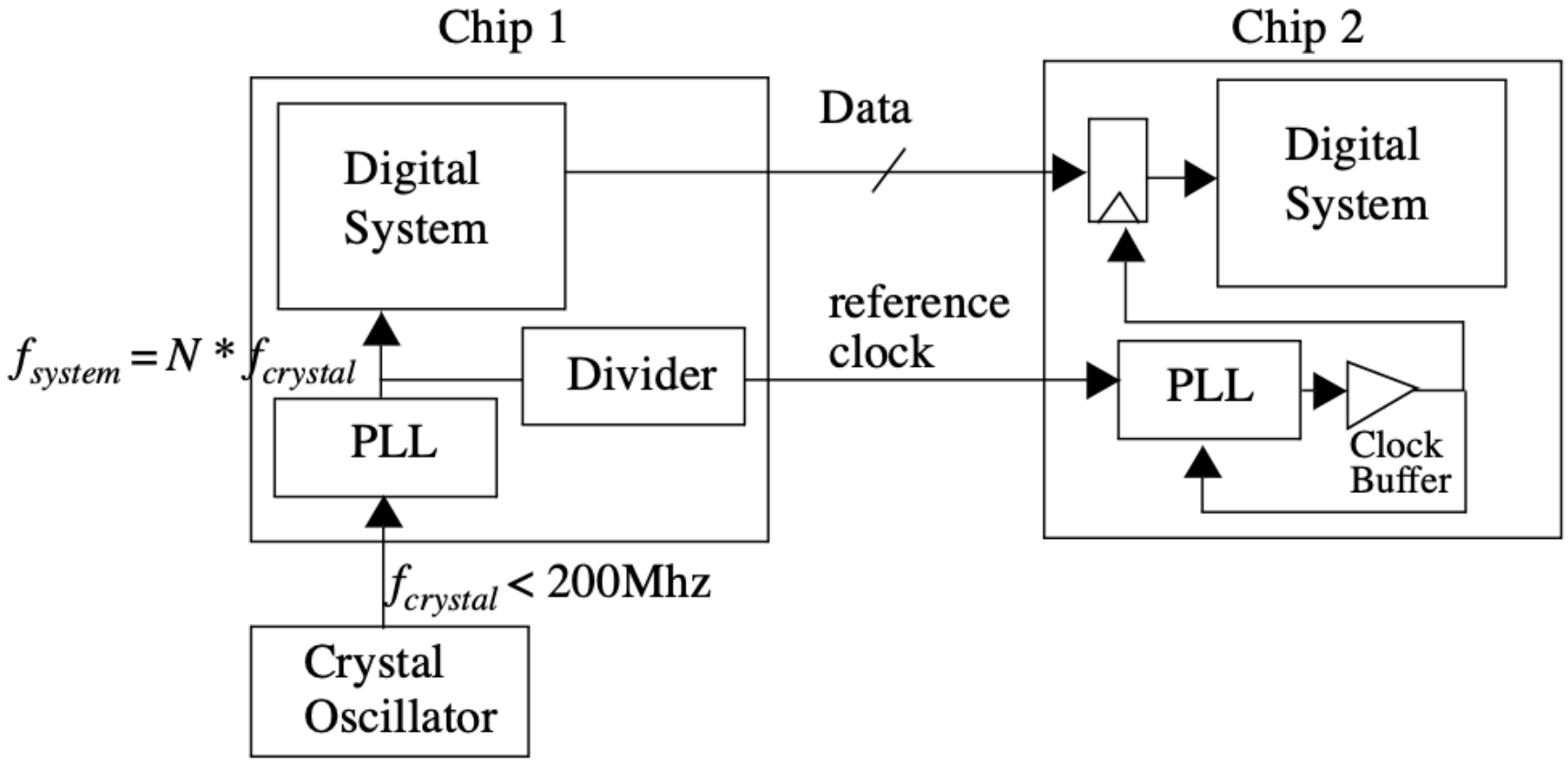
Other synchronizer options





00 → 0
01 → 1
11 → 2
10 → 3

We cheat and make them synchronous inputs



[Digital Integrated Circuits: A Design Perspective]

What’s the main exception here?

```
create_clock -name clk -period 500 [get_ports {clk}]

set_input_delay -clock [get_clocks clk] -min -add_delay 100 [all_inputs]
set_input_delay -clock [get_clocks clk] -max -add_delay 400 [all_inputs]

set_output_delay -clock [get_clocks clk] -min -add_delay 2 [all_outputs]
set_output_delay -clock [get_clocksclk] -max -add_delay  400 [all_outputs]
```

What's the main exception here? RESET

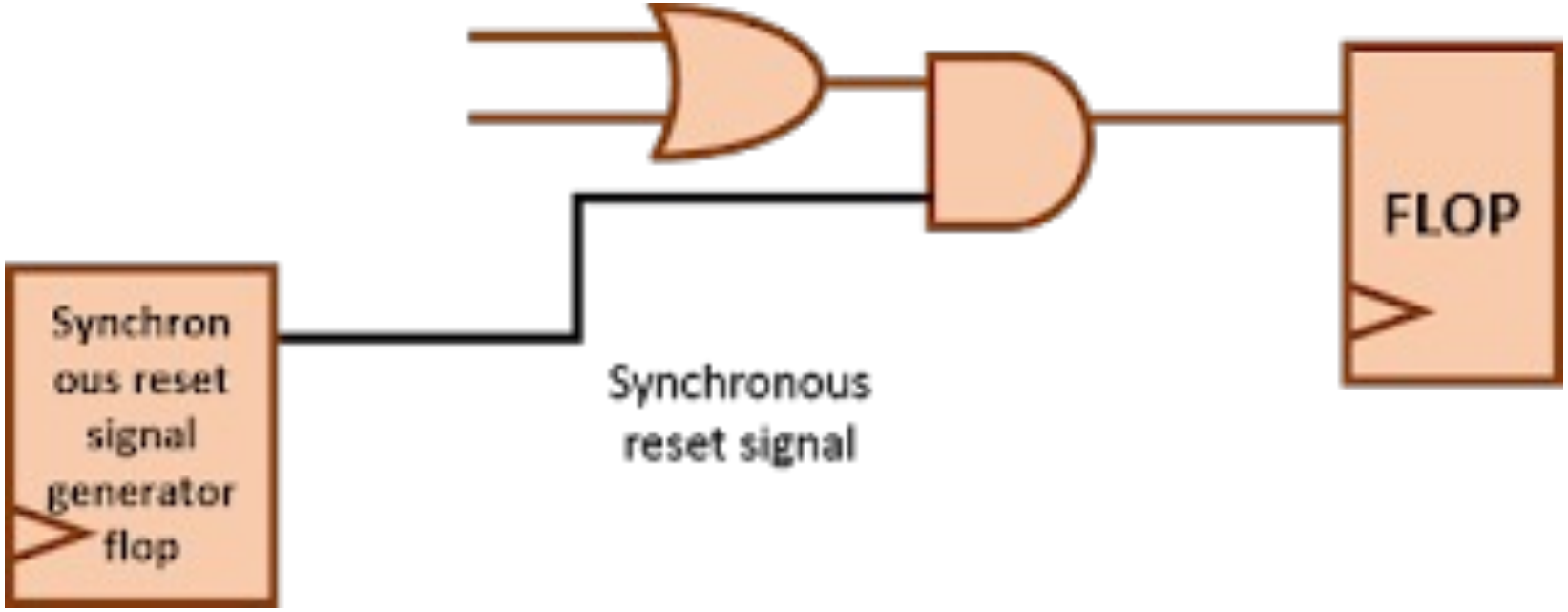
```
create_clock -name clk -period 500 [get_ports {clk}]

set_input_delay -clock [get_clocks clk] -min -add_delay 100 [all_inputs]
set_input_delay -clock [get_clocks clk] -max -add_delay 400 [all_inputs]

set_output_delay -clock [get_clocks clk] -min -add_delay 2 [all_outputs]
set_output_delay -clock [get_clocksclk] -max -add_delay  400 [all_outputs]
```

What makes RESET special?

- A RESET forces all sequential elements (like flip-flops) into a known, stable state, preventing undefined behavior during power-up or operation.
- Two ways to implement reset internally:
  - Asynchronous
  - Synchronous
- Where does reset come from?

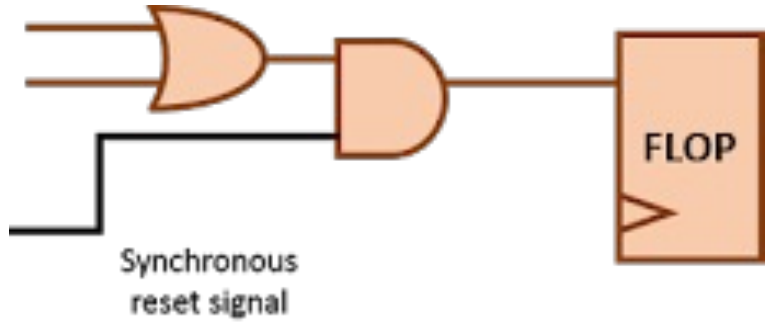
Sync Reset.

- Reset is synchronized to the clock
- Reset just another input
- Usually used for internal reset signals



https://vlsiuniverse.blogspot.com/2017/04/synchronous-and-asynchronous-resets.html

Sync Reset.

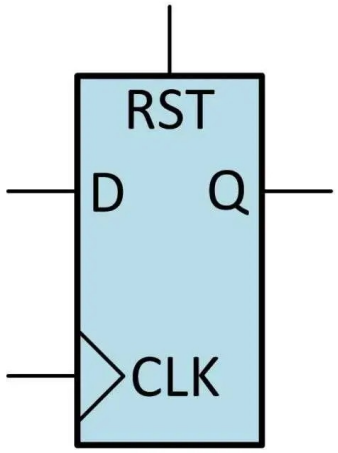


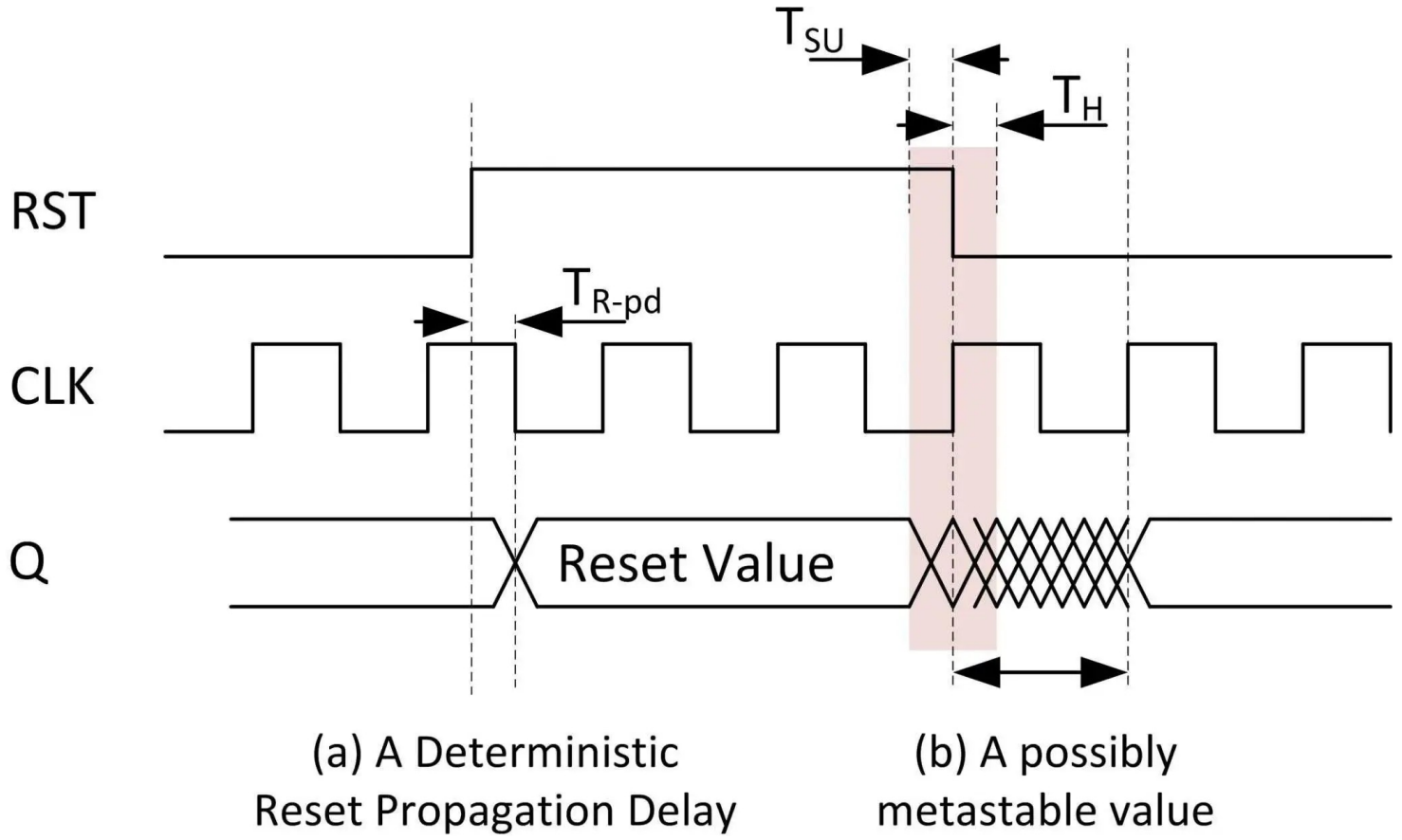

```
always_ff @(posedge clk) begin
    if(rst)
        q <= 1'h0;
    else
        q <= d;
end
```


https://vlsiuniverse.blogspot.com/2017/04/synchronous-and-asynchronous-resets.html

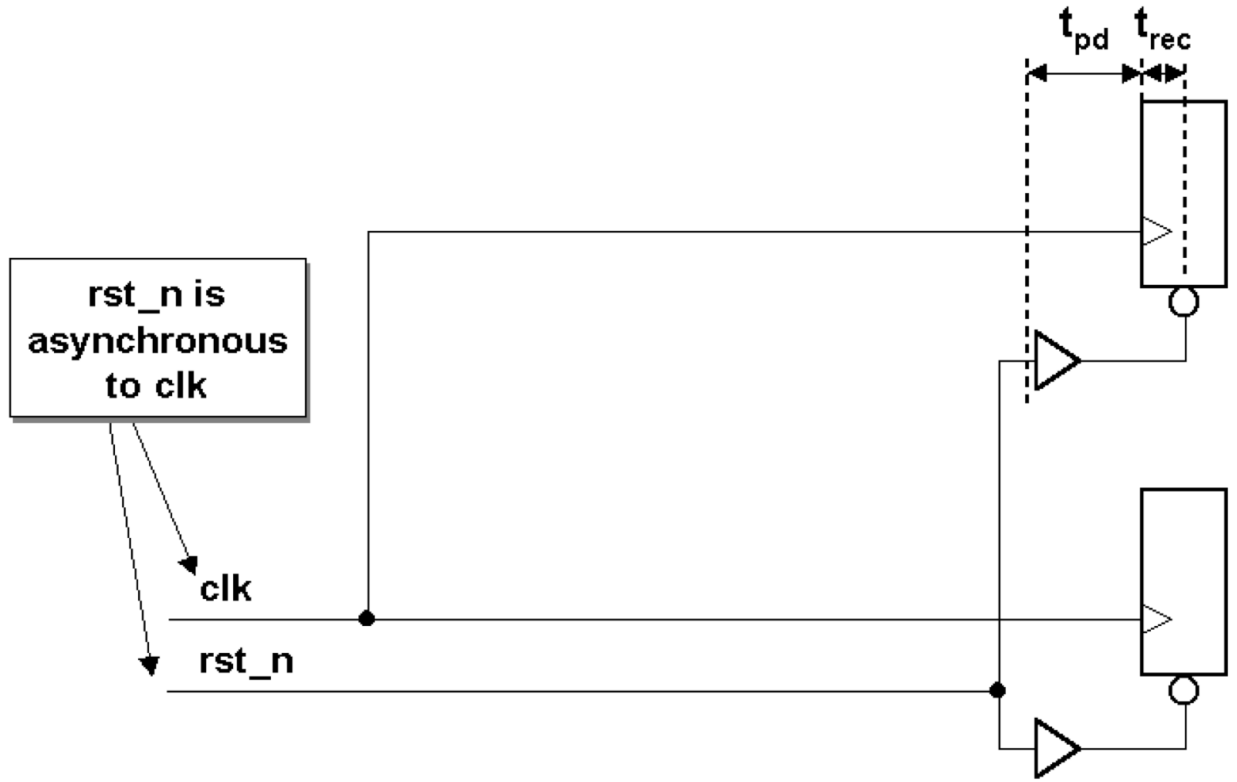
Async Reset.

- Reset is asynchronized to the clock
- Reset special input to flop
- Usually used for external reset signals



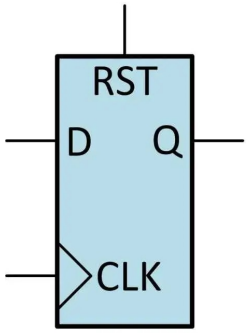


https://www.embedded.com/asynchronous-reset-synchronization-and-distribution-challenges-and-solutions/



http://www.sunburst-design.com/papers/CummingsSNUG2003Boston_Resets.pdf

Async Reset.



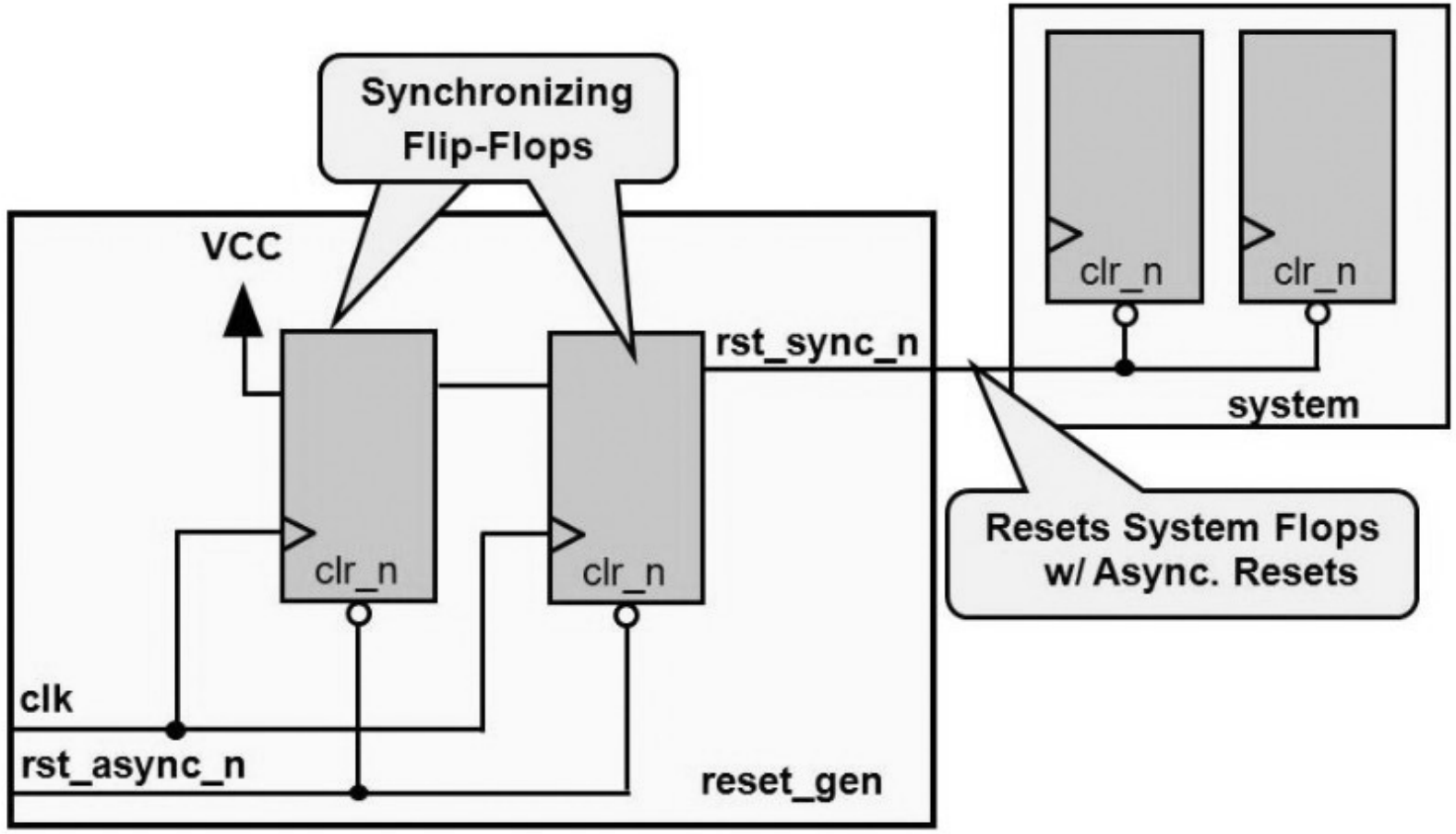

```
always @( posedge clk, posedge rst)
    if ( rst)
        q <= 'h0;
    else
        q <= d;
```


https://www.embedded.com/asynchronous-reset-synchronization-and-distribution-challenges-and-solutions/

Async Set / Sync Removal Reset

https://www.vemeko.com/blog/differences-between-asynchronous-and-synchronous-resets.html

Async Set / Sync Removal Reset



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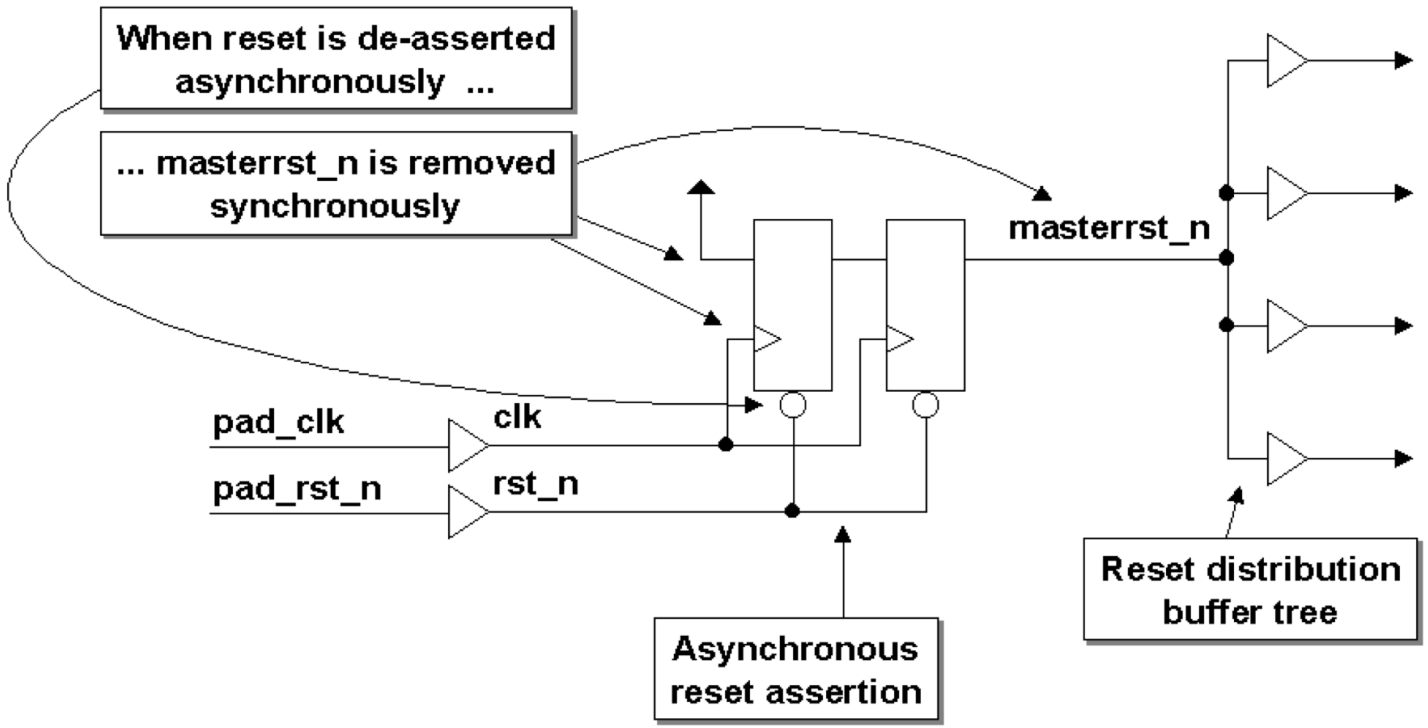

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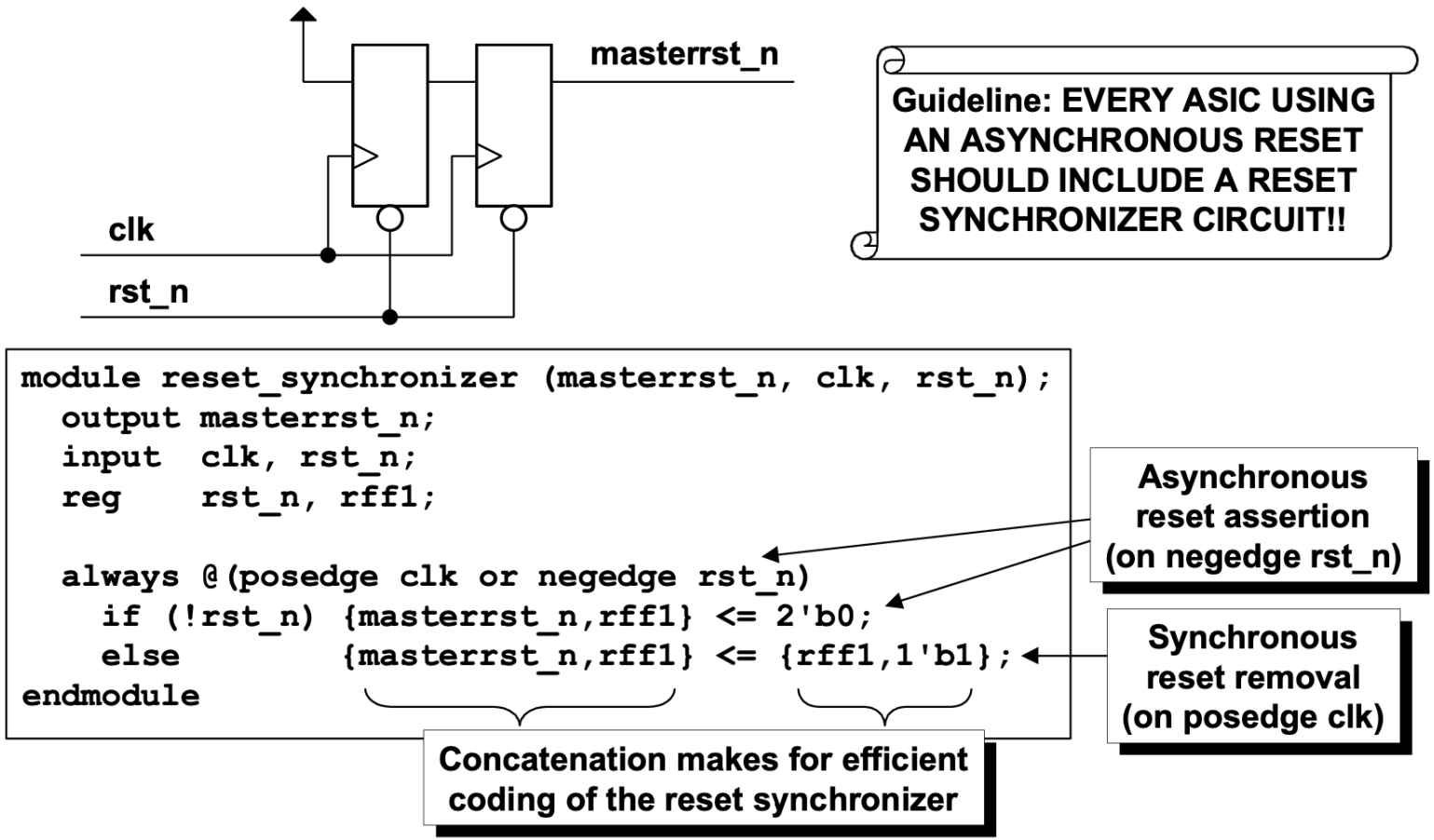
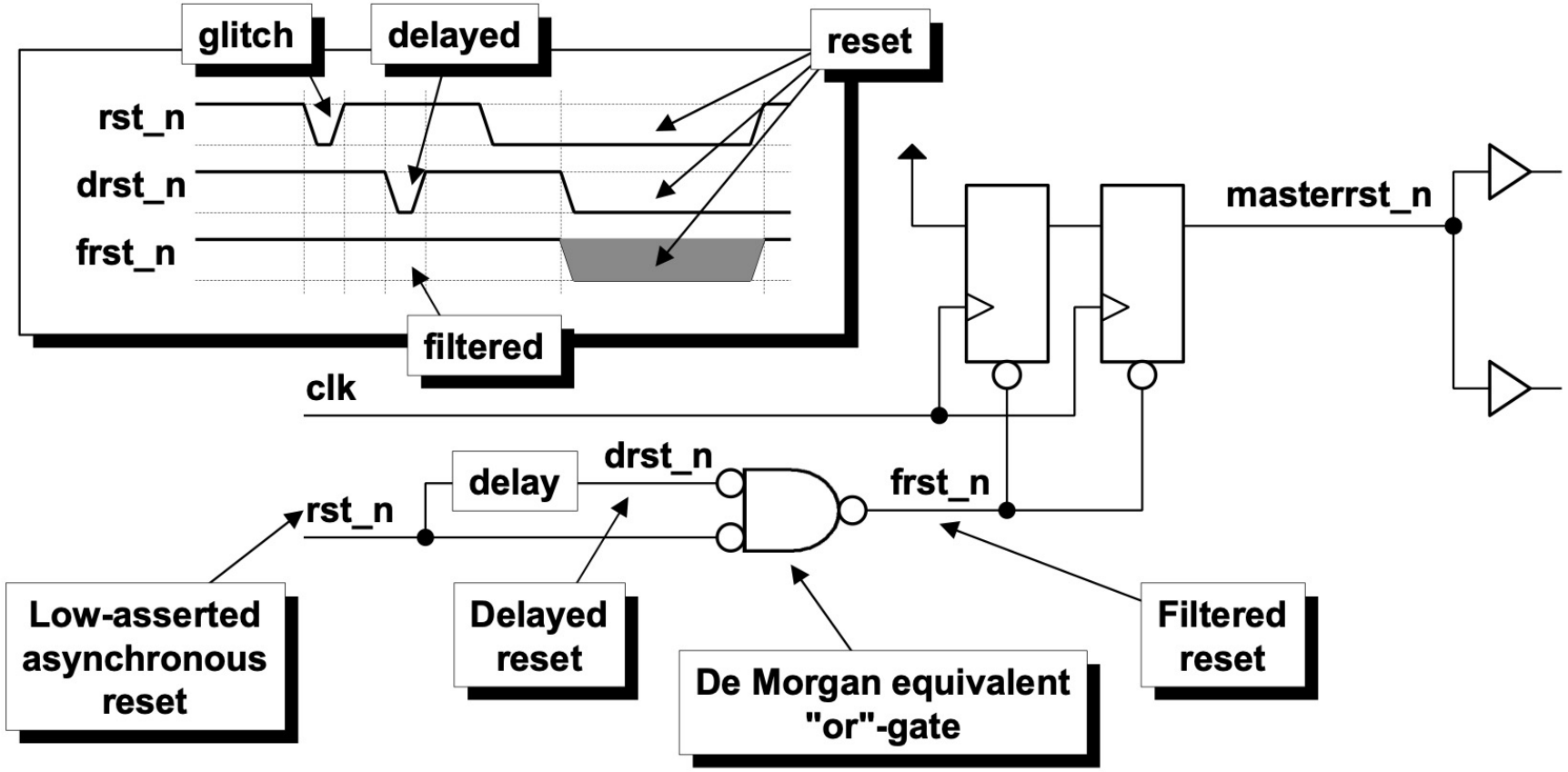


http://www.sunburst-design.com/papers/CummingsSNUG2003Boston_Resets.pdf

Stop

- Glitches on the rst_n input might cause stray resets
- Solution: glitch-filter on the rst_n input

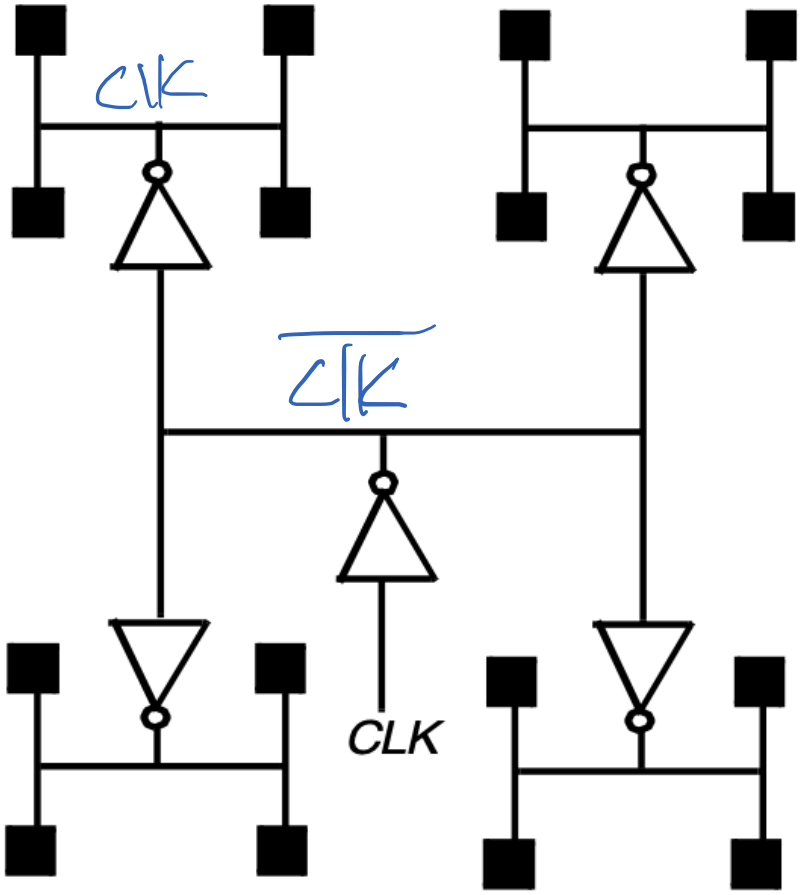


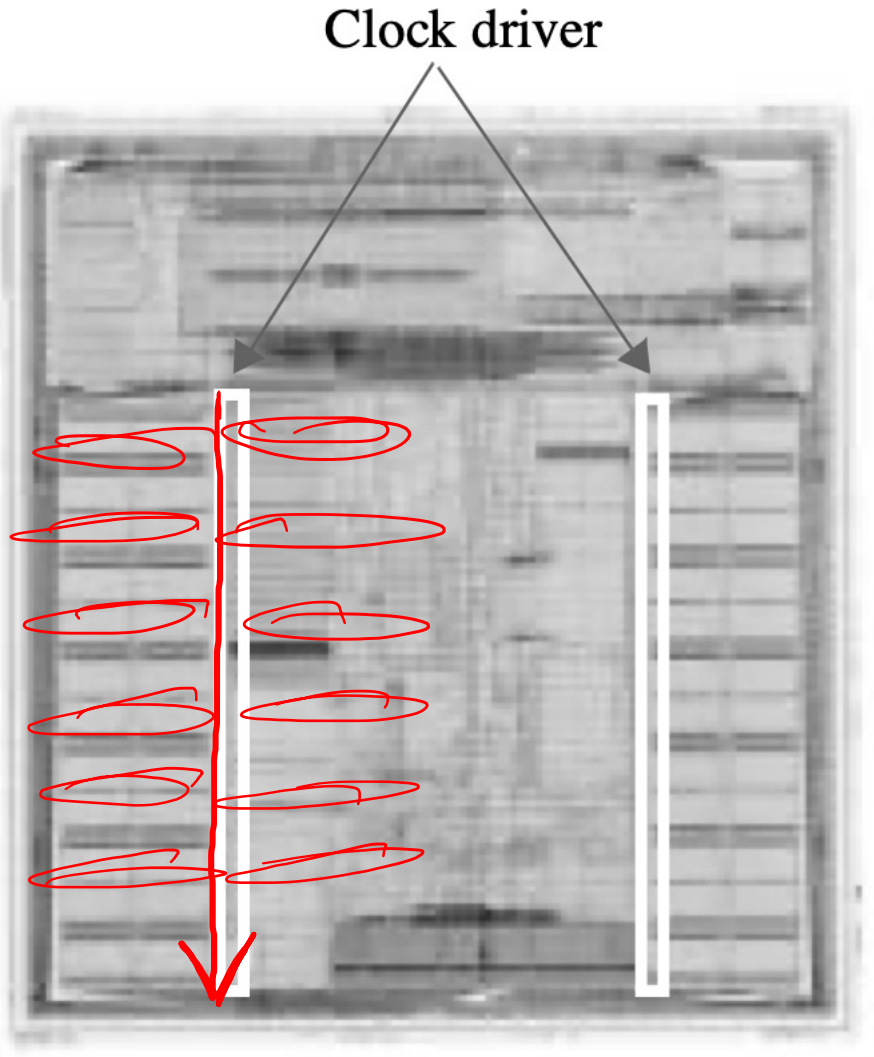
http://www.sunburst-design.com/papers/CummingsSNUG2003Boston_Resets.pdf

Clock Distribution to minimize skew

- H-Tree
- Fishbone

H-Tree vs. Fish-Bone





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